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Master Thesis

“Recurrence-Death Clustered-States Markovian Arrival Processes”

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SUMMARY

Counting processes are a class of mathematical objects which allow for different rich structures and model many real life processes. One of the first counting processes studied in history is the Poisson process, which obtained its name from the fact that any sample of points from the same time interval comes from a Poisson distribution. The Poisson process is tractable and found many real life applications, such as for the modeling of radioactive decay, but it fails to capture some complexities present in real life. The Poisson process cannot model non-identically distributed, dependent inter-event times, and it also cannot model non-exponentially distributed inter-event times. Thus, there was a need for investigating new counting processes.

In the 1970s, Neuts developed the Markovian Arrival Process (MAP), a counting process where state transitions within a continuous-time Markov chain may be recorded or not. This process can model non-identically distributed, non-exponentially distributed, dependent inter-event times; and in fact, these follow PH-Type distributions, which are matrix generalizations of the exponential distribution. However, MAPs only register events of one type. For example, this means that MAPs may not accurately model datasets with recurrence times for terminal-stage cancer patients who eventually pass away. In spite of this, the theory can be adapted to create better suited models.

The Clustered-States Markovian Arrival Processes (CS-MAPs) were proposed as a generalization of the MAP, where states are given group tags. In this thesis, our focus will be directed towards the Recurrence-Death CS-MAPs (RD CS-MAPs), where all states except for one are given a *recurrence* tag, while the other one is given a *death* tag, and the state with the *death* tag is absorbing.

Statistical inference for CS-MAPs is challenging because the loglikelihood function exhibits many local maxima. Moreover, RD CS-MAP correlations are usually low; and even though RD CS-MAPs with high correlations exist, these are hard to come by. Therefore, new techniques needed to be explored in order to fit RD CS-MAP models into real life datasets with high correlations.

In this Master's Thesis, the groundwork for understanding RD CS-MAPs is reviewed, beginning with some mathematical preliminaries, moving onto MAPs, and later mentioning general CS-MAPs and specifying RD CS-MAPs. Next, a review of the main theoretical properties of RD CS-MAPs is given, alongside a new formula for the moments of the number of recurrences, and then the results from two sampling experiments with

high and low inter-event time correlations are contrasted. Lastly, we explore the results obtained from applying our own optimizer, based on Scipy's `minimize` function with the `trust-const` basis on various kinds of datasets, from simulated samples to the real datasets `bladder1` and `colorectal`.

All of the Python code used for this thesis is publicly available within a GitHub repository. [If the reader is using a device with an Internet connection, they may click on this sentence to access the repository.](#)

Keywords: inter-event time, correlations, loglikelihood, optimization, censorship

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1. INTRODUCTION

In many disciplines outside of Mathematics, it is important to study lapses of time experienced by individuals between the start of an “experiment” and the observation of an “event” of interest, defined by a field expert. In some contexts, the event is terminal, and so will end the experiment for the individual in question; while in other contexts, events of a recurrent type will not end the experiment until the terminal event is observed. These event times and end results are often recorded within what we will refer to as “lifeline datasets”. Some examples of useful applications include: cancer patients’ lives in Medicine, component times to failure in Engineering, times to purchase in Economics, customer churn times in Marketing, among many others.

The subfield of Statistics which is concerned with the rigorous study of these datasets is known as Survival Analysis, which proposes non-parametric models and regression models for lifeline datasets. Survival Analysis is an extremely useful mathematical tool which has been continuously used by researchers over the years, ever since the immediate arrival of computers after its inception [1]. Its main appeal is that it allows researchers to overcome the problem of “censorship”, the process by which information about the lifeline of an individual is lost to the researcher. Censorship is intrinsic to lifeline datasets because in many instances, it may not be ethical or even possible to prolong an experiment for an extended period of time. Censorship may present itself in three different forms:

1. Right censorship - The process by which an individual abandons or is withdrawn from the dataset before experiencing the terminal event.
2. Left censorship - The process by which an individual enters the dataset late, after their associated starting time.
3. Interval censorship - The process by which an individual leaves the dataset and comes back again into the experiment before experiencing the terminal event.

The most used survival-analytic technique for dealing with right and left censorship is the Kaplan-Meier estimator, a mathematical formula highly revered by biomedical laboratory researchers [1]. In short, it can be shown that using the Kaplan-Meier estimator is equivalent to imputing missing data in a way that makes the most out of the present data [2]. This is critically useful because experiments that produce lifeline datasets are often restricted by tight budgets and low sample sizes.

Matrix models have been at the heart of Survival Analysis ever since their conception in the 1970s with Neuts’ “Markovian Arrival Processes” (MAPs). The MAP resulted from the necessity of more complex models that capture non-i.i.d., non-exponential inter-event

times, unlike the Poisson Process; and also from the formalization of Linear Algebra in the previous century. MAPs are a Markov Chain (an abstract random walker) whose stops are governed by independent Poisson processes and recorded by another counting process.

In this thesis, we will explore the `bladder1` and `colorectal` datasets, which are standard benchmarks for survival analysts. As it turns out, the MAP is a poor fit for these datasets for one main reason: the process does not record the recurrences of patients.

In the search of alternatives to Neuts' MAP for the modeling of lifeline datasets, the Clustered-States MAP (CS-MAP) is proposed by Lillo and Ramírez-Cobo, the advisors of this thesis. The key difference between the MAP and the CS-MAP is that, in the CS-MAP, Markov chain states are “clustered” together, according to some rule, which results in completely different distributions for the inter-event times. In the context of the cancer datasets, it is common to define just two clusters: the “death” cluster containing a single state (this cluster is the terminal event), and a “recurrence” cluster containing all of the other states. This way, we obtain the Recurrence-Death CS-MAP (RD CS-MAP), a particular case of the CS-MAP.

The distribution function of the inter-event times was first derived in [3], while in [4] advancements in both the theoretical aspects of statistical inference and in some of the computational aspects regarding optimization were made. However, RD CS-MAP parameter sets generally produce inter-event time correlations very close to 0, which are usually too different from the correlations present in real life datasets. Fortunately, parameter sets with high correlations do exist, so the goal of the optimization procedure should be to find them, while at the same time properly estimating other statistics such as the inter-event time means.

Finally, in this Master's thesis, we build on top of prior work, proving that it is possible to correctly model inter-event time correlations not only in simulated samples, but also in the previously mentioned real life datasets, via the maximization of a penalized loglikelihood function of the RD CS-MAP parameters. Our optimizer is built from scratch using SciPy's `minimize` function with the `trust-constr` basis. Furthermore, we independently derive mathematical formulas for the theoretical moments of the number of recurrences of the RD CS-MAP, and we perform some sampling experiments where we distinguish between low correlations parameter sets and high correlations parameter sets.

1.1. The `bladder1` dataset

The `bladder1` dataset is a standard benchmark for survival analysts, and is included in R's *survival* package [5]. It contains 118 bladder cancer patients from a clinical trial,

where the patients were classified into three groups based on the type of therapy they received. Analyses of this dataset increased in frequency since the publishing of a relevant paper by Wei et al. (1989) [6]. Three different datasets stemming from the same study exist in R's *survival* package, but only *bladder1* serves us because it contains the “lifelines” variables we need. The original *bladder1* dataset contains 295 instances of rows representing recurrences of the patients, ordered by patient ID first and then chronologically. The following variables from the dataset were the only ones relevant for our purposes:

- **id**: The identification number of each patient. We may have several rows associated to a single patient by this key.
- **recur**: The total number of recurrences from a patient.
- **start**: The start of a time interval, which is always a recurrence time.
- **stop**: The end of a time interval, which may be a recurrence, a death or a censorship.
- **status**: The status of the patient at the corresponding **stop**. A value of 0 means the patient is removed by censorship, a value of 1 means the patient is alive, a value of 2 means the patient is removed due to death from bladder cancer, and a value of 3 means the patient is removed due to death from anything else.
- **enum**: The event occurrence number, per **id**.

The dataset also contains the variables **treatment**, **number**, **size**, **rtumor**, **rsize**, but they are not relevant for our purposes.

Table 1.1. Example of an arbitrary lifelines dataset.

id	T1	T2	T3	N	last_state
1	1.1	0.8	-1	2	1
2	2.1	1	3	3	0
3	0.5	0.8	-1	2	0

Our goal was to put these datasets into the format from Table 1.1. To this end, we created the new dataset in the following way:

- We created a new dataframe, with the old **id** column but with only one instance per value.
- To calculate the number of inter-event times per patient, we counted the number of rows associated to the patient in the original dataset. A new column with the name “N” was created.

- As many time columns were created as the maximum of the **N** variable, which is 10, by the names of **T1**, **T2**, ..., **T10**. These columns were filled in with the corresponding **stop–start** values; and for the “**Tn**” values with no corresponding time intervals, we inserted values of -1.
- To calculate the final status of the patient, we looked at the **state** value of each of last row per **id** in the original dataset. A new column by the name of **last_state** with this information was added. **We considered final recurrence states as censorships**, since we assume all patients eventually pass away; and **we also didn’t distinguish between death from bladder cancer and death from other causes**. Therefore, a value of 0 means the patient was censored, and a value of 1 means the patient passed away.

This finished the preprocessing for obtaining our lifelines dataset of interest. The original dataset may be found in the GitHub repository by the name of `bladder1.csv`, and the preprocessed dataset by the name of `bladder1_preprocessed.csv`.

The new dataset contains 118 instances, of which 89 (so, 75.4%) are censored. We performed an analysis for each of the first three times only because only for these times we believe there is sufficient data to perform statistical analyses.

- **T1**: 61 recurrences, 17 deaths and 40 censorships. A total of 78 uncensored times.
- **T2**: 36 recurrences, 4 deaths and 21 censorships. A total of 40 uncensored times.
- **T3**: 26 recurrences, 2 deaths and 8 censorships. A total of 28 uncensored times.

Also, let $\overline{T_i}|_{N \geq i}$ denote the inter-event time mean of the i^{th} inter-event times, and let $r(T_i, T_j | N \geq j)$ denote the inter-event time correlation of the i^{th} and j^{th} inter-event times. The statistics regarding the uncensored samples are:

$$\begin{array}{ll} \overline{T_1}|_{N \geq 1} = 11.94937 & r(T_1, T_2 | N \geq 2) = -0.4011953 \\ \overline{T_2}|_{N \geq 2} = 7.72093 & r(T_1, T_3 | N \geq 3) = -0.2413463 \\ \overline{T_3}|_{N \geq 3} = 6.8 & r(T_2, T_3 | N \geq 3) = 0.4790927 \end{array}$$

As we can see, the values of the correlations are high overall, in absolute value.

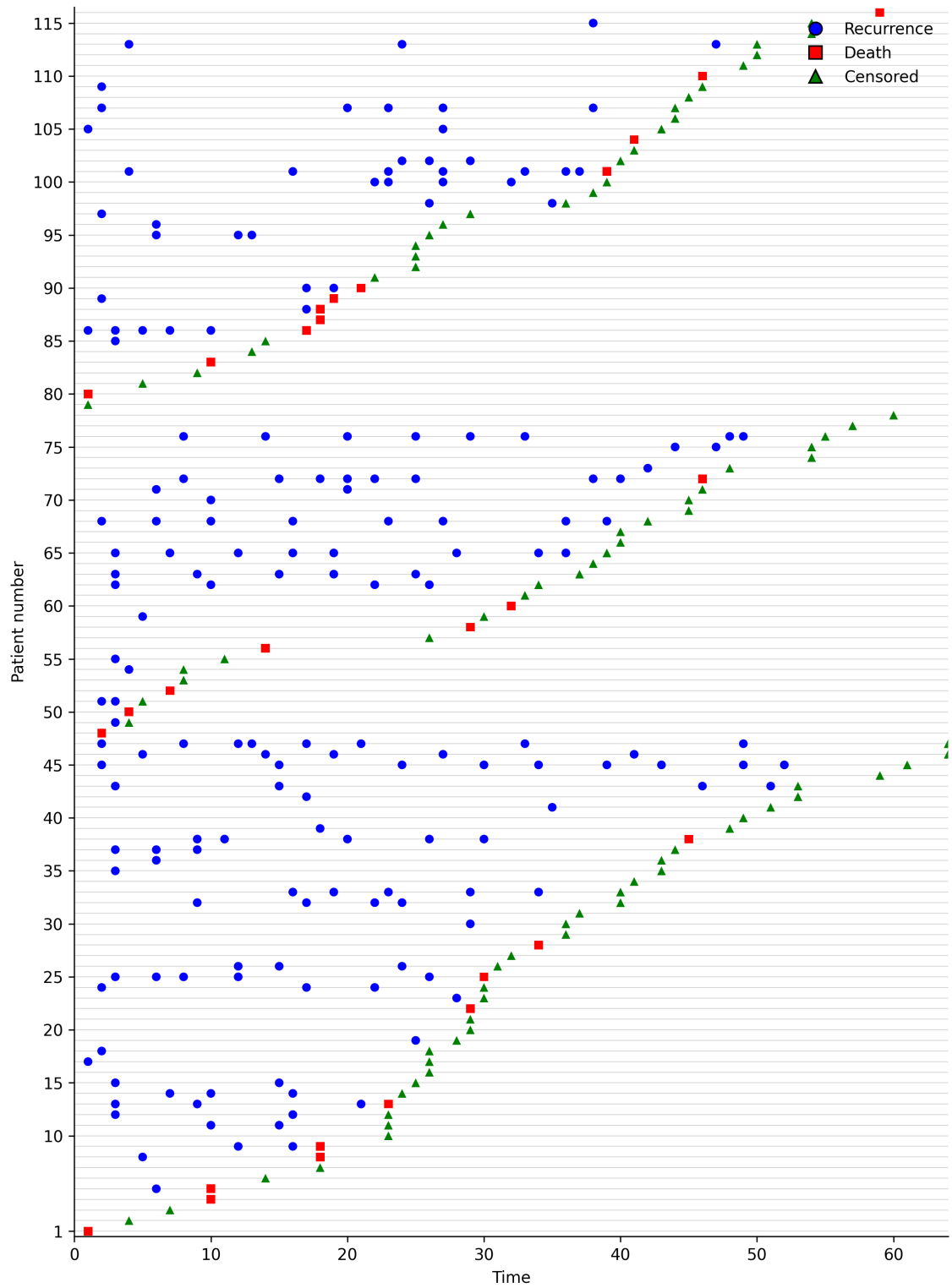


Fig. 1.1. An overview of the preprocessed lifelines bladder1 dataset.

1.2. The colorectal dataset

The colorectal dataset is another standard benchmark, included in R's *frailtypack* package [7]. It consists of 150 randomly chosen colorectal cancer patients from the FFCD

2000-05 multicenter phase III clinical trial, in which patients were classified into two groups based on the type of therapy they received. The original dataset contains 289 instances of rows representing event occurrences, ordered by patient ID first and then chronologically. The following variables from the dataset were the only ones relevant for our purposes:

- **id**: The identification number of each patient. We may have several rows associated to a single patient by this key.
- **time0**: Start of the time interval. Always a recurrence time.
- **time1**: End of the time interval. May be a recurrence, death or censoring time.
- **state**: Status of the patient at the corresponding **time1**. A value of 1 means the patient is in the dataset (alive), while a value of 0 means the patient is gone from the dataset (death or censorship).
- **gap.time**: The inter-event time.

The dataset also contains the variables **new.lesions**, **treatment**, **age**, **who.PS**, **prev.resection**, but we ignored them completely because we were only interested in the inter-event times and the status of the patients. It should also be noted that **gap.time** = **time1** – **time0**, meaning that there was some perfect multicollinearity within the dataset's variables. To reduce it, we removed the **time0** variable.

To put the dataset into the new format, we did the following:

- We created a new dataframe, with the old **id** column but with only one instance per value.
- To calculate the number of inter-event times per patient, we counted the number of rows associated to the patient in the original dataset. A new column with the name “**N**” was created.
- As many time columns were created as the maximum value of the **N** variable, which is 5, by the names of **T1**, **T2**, **T3**, **T4**, **T5**. These columns were filled in with the corresponding **gap.time** values; and for the “**Tn**” values with no corresponding **gap.time**, we inserted values of -1.
- To calculate the final status of the patient, we looked at the **state** value of each of last row per **id** in the original dataset. A new column by the name of **last_state** with this information was added.

This finished the preprocessing for obtaining our lifelines dataset of interest. The original dataset may be found in the GitHub repository by the name of `colorectal.csv`, and the

preprocessed dataset by the name of `colorectal_preprocessed.csv`.

The new dataset contains 150 instances, as was expected, of which 29 (so 19.3%) are censored. We describe all of the times here, but we only analyze the first three times in this thesis.

- **T1**: 80 recurrences, 41 deaths and 29 censorships. A total of 121 uncensored times.
- **T2**: 34 recurrences, 46 deaths and 8 censorships. A total of 72 uncensored times.
- **T3**: 11 recurrences, 23 deaths and 2 censorships. A total of 34 uncensored times.
- **T4**: 2 recurrences, 9 deaths and 0 censorships. A total of 11 uncensored times.
- **T5**: 0 recurrences, 2 deaths and 0 censorships. A total of 2 uncensored times.

Moreover, following the same notation as with `bladder1`, the statistics of interest for the uncensored values of this dataset are:

$$\begin{aligned}\overline{T_1}|_{N \geq 1} &= 0.7547053 & r(T_1, T_2 | N \geq 2) &= 0.35991192 \\ \overline{T_2}|_{N \geq 2} &= 0.5101443 & r(T_1, T_3 | N \geq 3) &= 0.07997016 \\ \overline{T_3}|_{N \geq 3} &= 0.379685 & r(T_2, T_3 | N \geq 3) &= 0.09345288\end{aligned}$$

As we can see, two of the correlations are small in absolute value, but the first one is not.

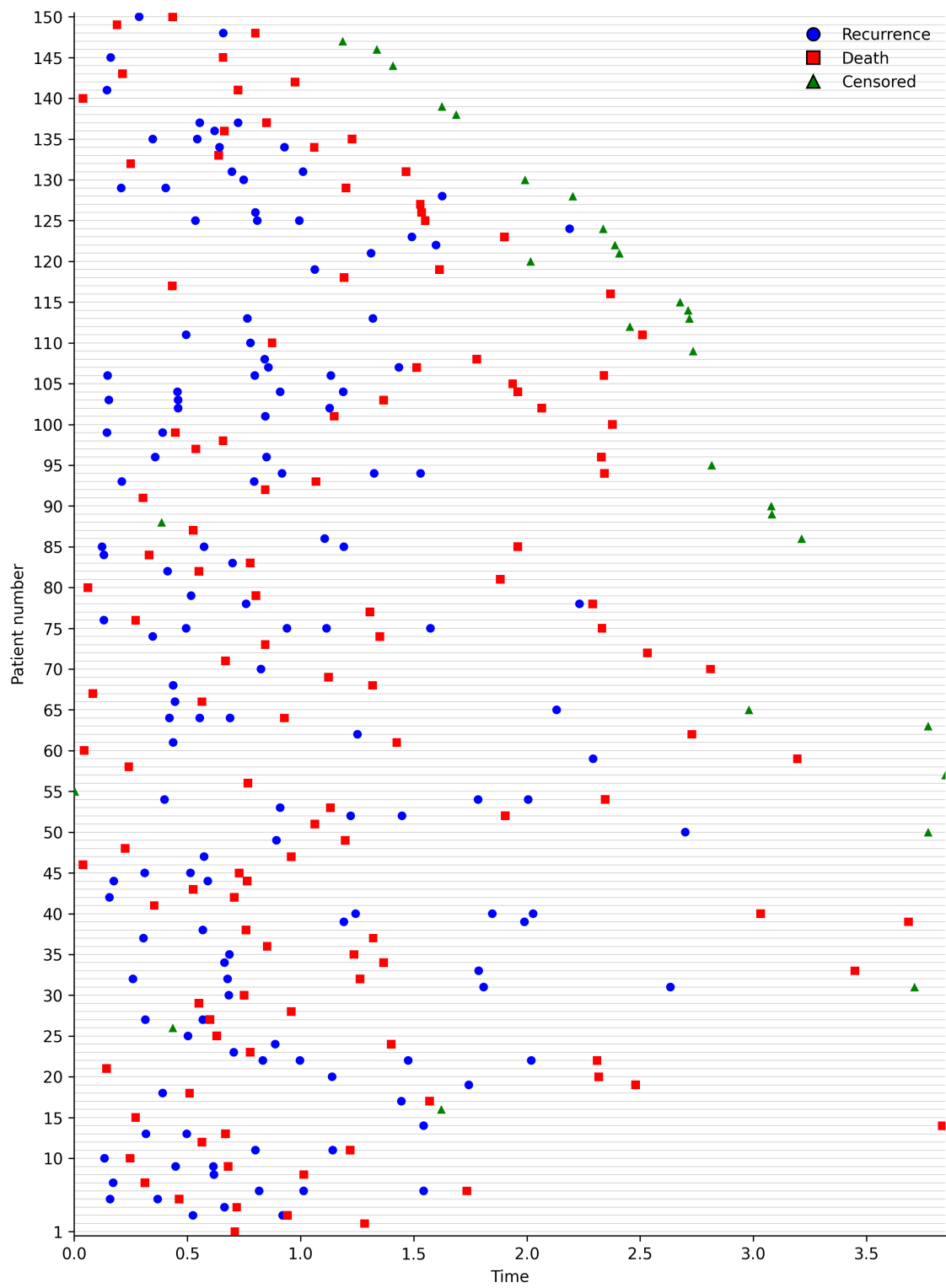


Fig. 1.2. An overview of the preprocessed lifelines colorectal dataset.

2. PRELIMINARIES

In order to understand the Markovian Arrival Process, some university-level mathematical concepts must be revised. The reader familiarized with the concepts of “Markov chains”, “Poisson processes” or “Survival Analysis” may skip sections within this chapter as they please.

2.1. Markov Chains

One of the main prerequisites needed to understand MAPs and CS-MAPs are Markov chains, mathematical objects which have always been of great relevance in Probability Theory and Mathematical Economics. Loosely speaking, Markov chains are the mathematical abstraction of a “random walker”, a person who randomly walks from some known locations into others.

Definition 1: A **stochastic process** is an infinite ordered set of random variables over a fixed probability space, $\{X_j | j \in J\}$. The index set J may be countably or uncountably infinite.

Definition 2: A **discrete-time Markov chain** (or process) is a countable stochastic process $\{X_n | n \in \mathbb{N}_{\geq 0}\}$ with values in a set $\mathcal{S}$ such that for all $n \in \mathbb{N}_{\geq 0}$ and $j \in \mathcal{S}$:

$$\mathbb{P}(X_{n+1} = j | X_0, \dots, X_n) = \mathbb{P}(X_{n+1} = j | X_n).$$

Definition 3: A **continuous-time Markov chain** (or process) is an uncountable stochastic process $\{X_t | t \geq 0\}$ with values in a set $\mathcal{S}$ such that for all $0 \leq s < t$ and $j \in \mathcal{S}$:

$$\mathbb{P}(X_t = j | \forall u \in [0, s]: X_u) = \mathbb{P}(X_t = j | X_s).$$

The defining equations described above are also known as the **Markov property**, and the set $\mathcal{S}$ is called the **state space**. Intuitively speaking, Markov chains are random processes whose future outcomes are determined only the most recent outcome, instead of the full history of outcomes.

Definition 4: A **discrete-time Markov chain** $\{X_n | n \in \mathbb{N}_{\geq 0}\}$ with state space $\mathcal{S}$ is said to be **time-homogeneous** if for all time instances $n \in \mathbb{N}_{\geq 0}$, and states $i, j \in \mathcal{S}$ we have:

$$\mathbb{P}(X_{n+1} = j | X_n = i) = \mathbb{P}(X_1 = j | X_0 = i).$$

Definition 5: A **continuous-time Markov chain** $\{X_t | t \geq 0\}$ with state space $\mathcal{S}$ is said to be **time-homogeneous** if for all time instances $0 \leq s < t$ and states $i, j \in \mathcal{S}$ we have:

$$\mathbb{P}(X_t = j | X_s = i) = \mathbb{P}(X_{t-s} = j | X_0 = i).$$

So, in other words, time-homogeneous Markov chains are Markov chains where the point at which the stochastic process starts is not important. From here onwards, we will assume all Markov chains to be time-homogeneous, unless otherwise stated. Moreover, we will also assume all Markov chains to have a finite set of states, composed of integers.

Some cornerstone concepts regarding Markov chains are the transition matrix, and the infinitesimal generator, for discrete-time and continuous-time chains respectively.

Definition 6: Let $\{X_n | n \in \mathbb{N}_{\geq 0}\}$ be a discrete-time Markov chain with state space $\mathcal{S}$. The n -step **transition matrix** of $\{X_n | n \in \mathbb{N}_{\geq 0}\}$ is defined to be the matrix $\mathbf{P}(n)$ with entries

$$p_{ij}(n) \equiv \mathbb{P}(X_n = j | X_0 = i).^1$$

Thus, the n -step transition matrix is the matrix containing in its $(i, j)^{\text{th}}$ entry the probability of the chain changing from state i to state j after n steps. Regarding discrete-time Markov chains, the Markov property immediately gives us the following characterization.

Proposition 1: Let $\{X_n | n \in \mathbb{N}\}$ be a discrete-time Markov chain, with 1-step transition matrix $\mathbf{P} \equiv \mathbf{P}(1)$. Then, for all $n \in \mathbb{N}$:

$$\mathbf{P}(n) = \mathbf{P}^n.$$

More detailed proofs can be checked in most intermediate Probability textbooks, by means of the ‘‘Chapman-Kolmogorov equations’’ [8, pages 194-196]. We have analogous results for continuous-time Markov chains.

Definition 7: Let $\{X_t | t \geq 0\}$ be a continuous-time Markov chain with state space $\mathcal{S}$. The value of the **transition function** at time $t \geq 0$ is defined to be the matrix $\mathbf{P}(t)$ with entries

$$p_{ij}(t) \equiv \mathbb{P}(X_t = j | X_0 = i).$$

It can be shown that transition functions are differentiable, allowing us to define generators [9].

Definition 8: Let $\{X_t | t \geq 0\}$ be a continuous-time Markov chain with transition function $\mathbf{P}$ and derivative $\mathbf{P}'$. The **(infinitesimal) generator** of $\{X_t | t \geq 0\}$ is defined to be the matrix $\mathbf{G} = \mathbf{P}'(0)$.

It may also be derived that the diagonal elements of generators are non-positive, while the off-diagonal elements are non-negative. Generators are useful because they allow us to calculate different transition function values without calculating each of the probabilities.

¹Note that the definition changes if the chain is not assumed to be time homogeneous. The equation would be replaced by $p_{ij}(n) \equiv \mathbb{P}(X_{m+n} = j | X_m = i)$ for all times $m \in \mathbb{N}_{\geq 0}$.

Proposition 2: Let $\{X_t | t \geq 0\}$ be a continuous-time Markov chain with transition function $\mathbf{P}$ and generator $\mathbf{G}$. Then for all $t \geq 0$:

$$\mathbf{P}(t) = e^{t\mathbf{G}} \equiv \sum_{k=0}^{\infty} \frac{1}{k!} (t\mathbf{G})^k.$$

2.2. Poisson Processes

In this section, the origins of counting processes are revised, starting from the classic Poisson process. The history of the Poisson process and the Poisson distribution spans several centuries, starting with the French mathematician Siméon Denis Poisson (1781-1840) himself, and receiving its modern definition by Soviet mathematicians and forefathers of axiomatic probability theory Andrey Kolmogorov (1903-1987) and Aleksandr Khinchin (1894-1959) (see [10] for a broad overview).

In order to construct the Poisson process from scratch, first we must define a more general type of stochastic process.

Definition 9: A *counting process* is a stochastic process $\{N_t | t \geq 0\}$ of non-negative, integer-valued random variables such that $\forall s \leq t: N_s \leq N_t$ (i.e. the process is non-decreasing).

Thus, a counting process represents the counting of event occurrences up to a certain time. Counting processes may also possess the following properties.

Definition 10: Let $\{N_t | t \geq 0\}$ be a counting process. We say that:

- The counting process has *independent time increments* if $\forall 0 \leq s_1 < s_2$ and $0 \leq t_1 < t_2$ with $t_1 \leq s_2$ we have that $N_{s_1} - N_{t_1}$ and $N_{s_2} - N_{t_2}$ are independent random variables.
- The counting process $\{N_t | t \geq 0\}$ has *stationary time increments* if $\forall 0 \leq s_1, s_2, t$ we have that N_{s_1+t} and N_{s_2+t} are identically distributed random variables.

So in other words, counting processes with independent time increments are such that the numbers of event occurrences within disjoint time intervals are independent; and counting processes with stationary increments are such that the numbers of event occurrences within intervals of the same length behave similarly (in a way, this last property is analogous to Markov chain time homogeneity).

Furthermore, if we ask for some standardization and slow growth properties, we obtain what is known as a Poisson process.

Definition 11: Let $\{N_t | t \geq 0\}$ be a counting process and $\lambda > 0$. We say that $\{N_t | t \geq 0\}$ is a *Poisson process with rate λ* if the following hold:

- $N_0 = 0$.
- $\{N_t | t \geq 0\}$ has independent and stationary time increments.
- For all $t \geq 0$:
 - $\mathbb{P}(N_t = 1) = \lambda t + o(t)$.
 - $\mathbb{P}(N_t > 1) = o(t)$.

The reader should also note that, from the previous definition it must follow that

$$- \mathbb{P}(N_t = 0) = 1 - \lambda t + o(t).$$

In short, it is desirable to control the growth of the counting process in the simplest but non-trivial mathematical way possible: with a first-order Taylor approximation. As it turned out, these properties combined are so strong that they define the counting process completely.

Proposition 3: (First characterization of the Poisson process)

Let $\{N_t | t \geq 0\}$ be a counting process and $\lambda > 0$ such that

- $N_0 = 0$.
- $\{N_t | t \geq 0\}$ has independent time increments.

Then, the following are equivalent:

- a) $\{N_t | t \geq 0\}$ is a Poisson process with rate λ , in the sense of Definition 10.
- b) For all $t \geq 0$: N_t follows a Poisson distribution of rate λt ,
(i.e. for all $k \in \mathbb{N} \cup \{0\}$: $\mathbb{P}(N_t = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}$).

A proof of the previous proposition may be found in many Probability textbooks, and despite not being short, it is elementary [11][12].

For this thesis, the most important property about Poisson processes is their sequences of inter-event times.

Definition 12: Let $\{N_t | t \geq 0\}$ be a counting process. We say that:

- a) For all $n \in \mathbb{N}$: $X_n = \inf\{t \geq 0 | N(t) = n\}$ is the n^{th} **arrival time** (and by convention, $X_0 = 0$).

b) For all $n \in \mathbb{N}$: $T_n = X_n - X_{n-1}$ is the n^{th} **inter-arrival time**.

We may also recover the arrival times from the inter-arrival times via $X_n = \sum_{i=1}^n T_i$ (a telescoping sum), and the counting process from the arrival times via $N(t) = \max\{n \in \mathbb{N} \mid X_n \leq t\}$ [13, page 230]. Thus, it follows that every counting process has a unique sequence of inter-event times associated to it. In the case of the Poisson process, much more is true.

Proposition 4: (Second characterization of the Poisson process)

Let $\{N_t \mid t \geq 0\}$ be a counting process and $\lambda > 0$. Then, $\{N_t \mid t \geq 0\}$ is a Poisson process of rate λ if and only if the sequence of inter-event times consists of independent random variables following an exponential distribution of rate λ .

A proof of this fact is also elementary, though technical in nature [13, pages 231-232]. Also, as a corollary, it follows that Poisson processes are examples of continuous-time Markov chains.

As we have seen, Poisson processes are models that assume some very restrictive assumptions that do not always hold in real life situations. For example, in the bladder1 and colorectal cancer datasets, there is not enough statistical evidence to suggest that inter-event times are independent, or exponentially distributed, so new counting processes need to be explored. However, the Poisson process was useful for constructing the basis of more sophisticated models, which generalize over the Poisson process.

2.3. Survival Analysis

In this last subsection, we present the most general mathematical notions from Survival Analysis, which we will need for tackling the censorship problem within our datasets. There are many introductive resources on this topic freely available [14][15][16].

2.3.1. The survival and hazard functions

In the simplest and most general setting, we have an individual who we assume is going to die, and we define their time until death as the non-negative random variable T . It is intuitive to begin studying the distribution of T via its probability distribution function; however, this function only returns the probability of the individual dying before a certain time. A more appealing reformulation lies in the complementary function, which tells us the probability of the individual *surviving* up to a certain time.

Definition 13: Let T be a non-negative random variable. The survival function of T is given by $S(t) = \mathbb{P}(T > t)$, for all $t \geq 0$.

It should be noted that some authors define the survival function with a partial inequality:

$S(t) = \mathbb{P}(T \geq t)$. If the variable T is continuous, it won't make a difference, but if the variable T is discrete, a convention should be followed.

Another analogous function which we could think of analyzing is the hazard function: the probability of an individual dying within an infinitesimal time interval, given that they have already survived past a certain time, relative to the size of the time interval.

Definition 14: Let T be a non-negative random variable. The hazard function of T is given by $h(t) = \lim_{h \rightarrow 0} \frac{\mathbb{P}(t < T \leq t+h | T > t)}{h}$ for all $t \geq 0$.

A fundamental result in Survival Analysis is the relationship between the survival function and the hazard function: one can be obtained from the other. Moreover, the proof is short, so it has been included in this text.

Proposition 5: Let T be a non-negative random variable, and let $S(t)$ and $h(t)$ be its survival and hazard functions, respectively. Then:

$$S(t) = e^{-\int_0^t h(s)ds}, \quad h(t) = -\frac{S'(t)}{S(t)}.$$

Proof.

For the second equation, we have:

$$\begin{aligned} h(t) &= \lim_{h \rightarrow 0} \frac{\mathbb{P}(t < T \leq t+h | T > t)}{h} = \lim_{h \rightarrow 0} \frac{\mathbb{P}(t < T \leq t+h)}{h\mathbb{P}(T > t)} = \frac{1}{\mathbb{P}(T > t)} \lim_{h \rightarrow 0} \frac{\mathbb{P}(T \leq t+h) - \mathbb{P}(T \leq t)}{(t+h) - t} \\ &= \frac{1}{S(t)} \lim_{h \rightarrow 0} \frac{(1 - \mathbb{P}(T > t+h)) - (1 - \mathbb{P}(T > t))}{(t+h) - t} = -\frac{1}{S(t)} \lim_{h \rightarrow 0} \frac{S(t+h) - S(t)}{(t+h) - t} = -\frac{S'(t)}{S(t)}. \end{aligned}$$

Lastly, we obtain the first equation by solving the differential equation (for all $t \geq 0$, we integrate over the interval $[0, t]$):

$$\begin{aligned} h(t) = -\frac{S'(t)}{S(t)} = -[\log(S(t))]' &\implies \int_0^t h(s)ds = -(\log(S(t)) - \log(S(0))) = -\log(S(t)) \\ &\implies S(t) = e^{-\int_0^t h(s)ds}. \end{aligned}$$

Even though the survival and the hazard functions are completely determined by one another, having both was necessary for the development of Survival Analysis - it just happens that some problems are easier to tackle with one function over the other. In our case, the survival function proved to be more useful, but the hazard function was needed to prove the validity of our methods for handling censorship.

2.3.2. Censorship and the Kaplan-Meier estimator

From a Probability setting, the survival function completely determines the random life-lines of individuals from a sufficiently big population. When we perform Statistics, all we

can obtain are empirical statistics to this survival function, and perhaps the most intuitive of them all is the empirical survival function: the proportion of alive individuals with respect to the total number of individuals.

Definition 15: Let T be a non-negative random variable and let $T_1, \dots, T_n$ be an i.i.d. sample following the distribution of T . For all $t \geq 0$, the empirical survival function of the sample $T_1, \dots, T_n$ is defined as

$$\hat{S}(t) = \frac{|\{\text{Individuals alive at time } t\}|}{|\{\text{Individuals alive at time } 0\}|}.$$

The reader should note that empirical survival functions are always step functions, since they are based on the proportion of alive individuals and only finitely many individuals die, at times $0 < t_1, \dots, t_n$. At these precise times, individuals who will die are considered dead, by convention, which implies that empirical survival functions are right-continuous. Also note, an assumption that will always be made in survival-analytic models is that survival probabilities are constant over time.

In the presence of censored data, the empirical survival function severely underestimates the survival times of individuals because it treats censored individuals as “dead” individuals, since they have been removed from the dataset. In order to correct for censorship, we must assume the following three hypotheses about it: censorship is random (non-informative), censoring times are independent, and no other terminal events exist [17, Kaplan-Meier curve assumptions].

Satisfying all prior assumptions, Edward L. Kaplan (1920-2006) and Paul Meier (1924-2011) devised a new estimator to the survival function in their famous paper from 1957 [18].

Definition 16: Let T be a non-negative random variable and let $T_1, \dots, T_n$ be an i.i.d. sample following the distribution of T . Moreover, let $0 < t_1 < \dots, < t_k$ be all of the times where at least one event happened¹. Then, for all $t \geq 0$, the *Kaplan-Meier estimator* of the sample $T_1, \dots, T_n$ is defined as

$$\hat{S}(t) = \prod_{\{i | t_i \leq t\}} \left(1 - \frac{|\{\text{Deaths at time } t_i\}|}{|\{\text{Individuals alive at time } t_i\}|} \right).$$

In the case where there is no censorship within the data, $k = n$ and the Kaplan-Meier estimator reduces to the empirical survival function; thus, the Kaplan-Meier estimator generalizes over the empirical survival function to more complex lifeline datasets, such as those observed in real life scenarios. Moreover, the Kaplan-Meier estimator satisfies a very important statistical property: it is the maximum likelihood estimator of the survival function in both discrete [19] and continuous [15, pp. 9–8] models (even though the

¹Note: censorship is not an event.

details are not checked in this thesis, hazard functions are needed for proving this fact). Furthermore, the Kaplan-Meier estimator can be shown to be asymptotically consistent, unbiased, and even efficient (minimizing variance within the class of unbiased estimators), which follows from it being a maximum likelihood estimator[2, page 1408].

To summarize, the Kaplan-Meier estimator is a mathematically rigorous and well-established way of estimating the survival function of a lifelines population. Moreover, it is non-parametric, meaning that it doesn't introduce external biases into the estimation process.

2.3.3. Lifeline data imputation using the Kaplan-Meier estimator

To finish this section on Survival Analysis, the process of imputing data using the Kaplan-Meier estimator is detailed. The objective is to transform datasets with cases of censorship into datasets where there is no censorship. This is important for us because it allows for streamlining code, and it provides a much easier way of understanding the Kaplan-Meier estimator, setting aside mathematical complexities. Not only that, but for the purposes of handling censorship, data imputation is a more important task than visualizing survival function plots. For a thorough review and extension of this topic, the article by Stephen R. Cole et al. [2] should be consulted, as it was the main source used for this subsection.

In the article, right-censored individuals are assigned “masses”, based on how early they were censored and the step sizes of the estimator. These masses are then distributed into future times via “Efron’s redistribution algorithm” (cited in the original source), which is essentially a form of fairly distributing the masses based on the previous criteria. The assumption that censorship is random (non-informative) is crucial: it allows the algorithm to justify the proportional distribution of the mass of the censored individual into all event times after the censoring time.

This is best understood by working through a simple example.

Example 1: Consider the following dataset:

Table 2.1. Example lifelines dataset for Kaplan-Meier imputation, first stage.

Individual ID	Time	Status
1	4	1 (Dead)
2	7	1 (Dead)
3	10	1 (Dead)
4	5	0 (Censored)
5	8	0 (Censored)

Our goal is replace the censoring times of individuals 4 and 5 for some death times, using the Kaplan-Meier estimator. First, we design a spreadsheet which handles all of the necessary calculations related to the estimator's step sizes:

Table 2.2. First spreadsheet of the example lifelines dataset for Kaplan-Meier imputation, second stage.

Event time	N° alive	N° events	KM value	Step size
t_i	n_i	d_i	$\hat{S}(t_i)$	$\hat{S}(t_{i-1}) - \hat{S}(t_i)$
4	5	1	$1 - \frac{1}{5} = 0.8$	$1 - 0.8 = 0.2$
7	3	1	$(1 - \frac{1}{5})(1 - \frac{1}{3}) = 0.5\bar{3}$	$0.8 - 0.5\bar{3} = 0.2\bar{6}$
10	1	1	0	$0.5\bar{3} - 0 = 0.5\bar{3}$

Afterwards, we design another spreadsheet, with rows associated to censored individuals, which contain the censoring times, the future Kaplan-Meier estimator jumps (“future”, relative to this censored individual) and the proportions of these jumps.

Table 2.3. Second spreadsheet of the example lifelines dataset for Kaplan-Meier imputation, third stage.

ID	Censoring time	KM value	Future steps	Proportions
-	c_i	$\hat{S}(c_i)$	$(t_j, \hat{S}(t_{j-1}) - \hat{S}(t_j))$	$\frac{\hat{S}(t_{j-1}) - \hat{S}(t_j)}{\sum_{t_i > c_i} \hat{S}(t_{i-1}) - \hat{S}(t_i)}$
4	5	0.8	$(7, 0.2\bar{6}), (10, 0.5\bar{3})$	$\frac{0.2\bar{6}}{0.2\bar{6} + 0.5\bar{3}} = 0.3, \frac{0.5\bar{3}}{0.2\bar{6} + 0.5\bar{3}} = 0.6$
5	8	0.5 $\bar{3}$	$(10, 0.5\bar{3})$	1

With this, we can finally estimate each missing event time by calculating the associated weighted sums, or averages¹. Each “proportion” can be thought of as the probability of an event occurrence in the future, and quantities are the original event times, present in the dataset.

Table 2.4. Example lifelines dataset for Kaplan-Meier imputation, final stage.

Individual ID	Time	Status
1	4	1 (Dead)
2	7	1 (Dead)
3	10	1 (Dead)
4	$0.3 \cdot 7 + 0.6 \cdot 10 = 9$	1 (Dead, imputed)
5	$1 \cdot 10 = 10$	1 (Dead, imputed)

¹This last step is not featured in the original paper.

One important caveat is that the previously described procedure cannot impute times for individuals with censoring times greater than or equal to the highest event time. These individuals are very important to the dataset, since they represent the extremes of the population, and so should not be removed, even though we lack rigorous methods for imputing their times. However, in the context of CS-MAPs and the `bladder1` and `colorectal` datasets, we will have multiple times associated to a given extreme individual, and we will use them to our advantage to leverage reasonable estimations for the censored times. The idea is that, since extreme individuals deviate from the mean individual and from most other people, only the past records of these extreme individuals could serve for a reasonable estimation.

As it turns out, the `bladder1` dataset features censored instances of this kind, while the `colorectal` dataset does not.

3. THE MARKOVIAN ARRIVAL PROCESS

Next, a recap on the literature of Markovian Arrival Processes (MAPs) is given. These are much more recent models that generalize over the Poisson process, allowing for non-i.i.d., non-exponentially distributed inter-event times. An extensive review from [20] and [21] (Chapter 2) is recommended.

A MAP is a stochastic process which is defined by two elements: a latent continuous-time Markov chain, and a counting process defined casewise for each state of the Markov chain. If the continuous-time Markov chain has m states, then we say it is governed by $2m^2 - m = m(2m - 1)$ independent Poisson processes, because for each of the m states, $m - 1$ Poisson processes are associated to *transitions without arrivals*, while some other m Poisson processes are associated to *transitions with arrivals*. The difference between transitions with or without arrivals lies in the counting process: the counting process increases its count each time a transition with an arrival takes place. Therefore, in the context of MAPs, an arrival is defined to be an increase in the counting process, and they are used for modeling non-terminal events.

Below we present the definition of a MAP in more rigorous terms.

Definition 17: Let $m \in \mathbb{N}$, $\alpha_0 \in \mathbb{R}^m$ be a probability vector, and $D_0, D_1 \in \mathbb{R}^{m \times m}$ where D_0 is invertible, the diagonal elements of D_0 are negative, and the off-diagonal elements of D_0 and all elements of D_1 are non-negative. Denote the $(i, j)^{\text{th}}$ elements of the previous matrices by $D_0 \equiv (d^{(0)})_{ij}$ and $D_1 \equiv (d^{(1)})_{ij}$. We define the following:

- A continuous-time Markov chain $\{I(t) | t \geq 0\}$ defined by the infinitesimal generator $D_0 + D_1$ and the initial probability distribution α_0 (that is, the random distribution of $I(0)$ is governed by α_0).
- $2m^2 - m = m(2m - 1)$ Poisson processes with rates $d_{ij}^{(0)}$ for $1 \leq i, j \leq m$ with $i \neq j$ and $d_{ij}^{(1)}$ for $1 \leq i, j \leq m$ (if any of these rates is 0, the Poisson process is said to be degenerate and produces no events).
- A counting process $\{N(t) | t \geq 0\}$ such that:
 - $N(0) = 0$.
 - For all state pairs $(i, j) \in \{1, \dots, m\}^2$ and times $t > 0$: $N(t)$ increases by 1 whenever $I(t) = i$ and a transition from the Poisson process with rate $d_{ij}^{(1)}$ is made, whereas it does not increase whenever a transition from the Poisson process with rate $d_{ij}^{(0)}$ (or $-d_{ii}^{(0)}$ if $i = j$) is made.

Then, the stochastic process $\{(N(t), I(t)) | t \geq 0\}$ is called a MAP_m , and the matrices (α_0, D_0, D_1) are the *matrix representation* of the MAP_m .

Before showing examples, some important remarks must be made:

1. An alternative way of representing MAP parameters is via the *rates* vector λ and the *transition matrices* P_0, P_1 . The vector λ contains all of the rates of the Poisson process inter-event times, while P_0 is the transition matrix of transitions without arrivals and P_1 is the transition matrix of transitions with arrivals. These matrices also satisfy the following: $p_{ii}^{(0)} = 0$ for all $i \in \{1, \dots, m\}$, and $(P_0 + P_1)\mathbb{1} = 1$. Therefore, considering the $2m^2 + 2m$ total elements, the m linear restrictions from $P_0 + P_1$, the m diagonal elements restrictions from P_0 , and the probability vector restriction of α_0 introduces $2m+1$ linear restrictions into our parameter set, reducing our degrees of freedom to $2m^2 - 1$, just as before. The formulas that relate these matrices are:

$$\begin{aligned}\lambda &= (\lambda_i)_{i \in \{1, \dots, m\}} = (-d_{ii}^{(0)})_{i \in \{1, \dots, m\}} \\ P_0 &= \text{diag}(\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_m})D_0 + I, & D_0 &= \text{diag}(\lambda_1, \dots, \lambda_m)(P_0 - I) \\ P_1 &= \text{diag}(\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_m})D_1, & D_1 &= \text{diag}(\lambda_1, \dots, \lambda_m)P_1\end{aligned}$$

2. In order for $D_0 + D_1$ to be a generator, it must be the case that $(D_0 + D_1)\mathbb{1} = 0$ (i.e. row sums of $D_0 + D_1$ sum to 0). Thus, the diagonal of D_0 is completely defined by all other matrix entries of the same row of D_0 and D_1 :

$$d_{ii}^{(0)} = -\left(\sum_{j=1, j \neq i}^m d_{ij}^{(0)} + \sum_{j=1}^m d_{ij}^{(1)} \right).$$

3. The invertibility of D_0 is a very mild condition. For it to be satisfied, rather than imposed as a hypothesis, it is sufficient that D_1 has either no rows or no columns of all zeros. We provide this result as a theorem, whose proof can be found in the Appendix.

Theorem 1: Let (α_0, D_0, D_1) be the matrix representation of a MAP, with a not necessarily invertible matrix D_0 . If D_1 contains either no rows or no columns of all zeroes, then D_0 is invertible.

It is reasonable to assume that MAPs do not contain states whose to and from transitions never produce arrivals, states which are completely invisible to the counting process, forming a “cross” of zeroes within D_1 . However, since we are defining MAPs in full generality, the invertibility condition of D_0 must be enforced.

4. Every continuous-time Markov chain has an associated discrete-time Markov chain known as the *embedded Markov chain* (of the continuous-time Markov chain). It is naturally defined by the discrete set of times for which transitions of the continuous-time Markov chain occur. In the case of the MAP, the following proposition holds.

Proposition 6: Let (α_0, D_0, D_1) be the matrix representation of the MAP $\{(I(t), N(t)) | t \geq 0\}$. Then, the transition matrix of the embedded Markov chain of $I(t)$ is given by $P^* = -D_0^{-1}D_1$. In particular P^* is a row-stochastic matrix (i.e. such that $0 \leq (p^*)_{ij} \leq 1$ and $P^*\mathbb{1} = \mathbb{1}$).

5. In practice, when performing Statistics, we are focused on studying the counting process, rather than the continuous-time Markov chain, since this is usually unobserved. As such, in an abuse of language, MAPs are sometimes associated only to their counting processes.
6. Inter-event times of MAPs are not uniquely defined in the sense of the non-identifiability problem: for each MAP, there exists another MAP such that the inter-event times joint distributions of both associated MAPs are equal. This result was proven for a generalization of the MAP, the Batch MAP (BMAP), in [22] and [23]. This highlights the importance of studying the matrix representations of MAPs instead of MAP Markov chains.

We shall now see now examples of MAPs.

Example 2: (Poisson processes)

Every Poisson process $\{N(t) | t \geq 0\}$ of rate $\lambda \geq 0$ can be seen as a MAP_m for all $m \in \mathbb{N}$: we consider the continuous-time Markov chain $\{I(t) | t \geq 0\}$ given by the infinitesimal generator

$$\begin{bmatrix} -\lambda & \lambda & 0 & \dots \\ 0 & -\lambda & \lambda & \dots \\ 0 & 0 & -\lambda & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix},$$

which gives us the $\text{MAP}_m \{(N(t), I(t)) | t \geq 0\}$.

In the case of the MAP_1 , a matrix representation is given by $\alpha_0 = 1$, $D_0 = -\lambda$, and $D_1 = \lambda$. As for the converse, every MAP_1 with only transitions with arrivals can be seen as a Poisson process, by extracting the counting process and dropping the Markov chain; however, MAP_1 s are generally slightly more complex than Poisson processes.

Example 3: (2-dimensional example)

We consider the point process $N(t)$ and continuous-time Markov chain $I(t)$ given by

$$N(t) = \begin{cases} 0, & \text{if } 0 \leq t < 0.22 \\ 1, & \text{if } 0.22 \leq t < 0.75 \\ 2, & \text{if } 0.75 \leq t < 2.49 \\ 3, & \text{if } 0.49 \leq t < 2.69 \\ 4, & \text{if } 2.69 \leq t < 3 \end{cases}$$

$$I(t) = \begin{cases} 0, & \text{if } t \in [0.04, 0.13) \cup [0.22, 0.64) \cup [0.75, 2.05) \cup [2.06, 2.49) \cup [2.69, 3) \\ 1, & \text{if } t \in [0, 0.04) \cup [0.13, 0.22) \cup [0.64, 0.75) \cup [2.05, 2.06) \cup [2.49, 2.69) \end{cases}$$

These are the first 3 units of time of a MAP run, simulated from the parameter set

$$\alpha_0 = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix}, \quad D_0 = \begin{bmatrix} -2.5 & 2 \\ 3 & -10 \end{bmatrix}, \quad D_1 = \begin{bmatrix} 0 & 0.5 \\ 7 & 0 \end{bmatrix}.$$

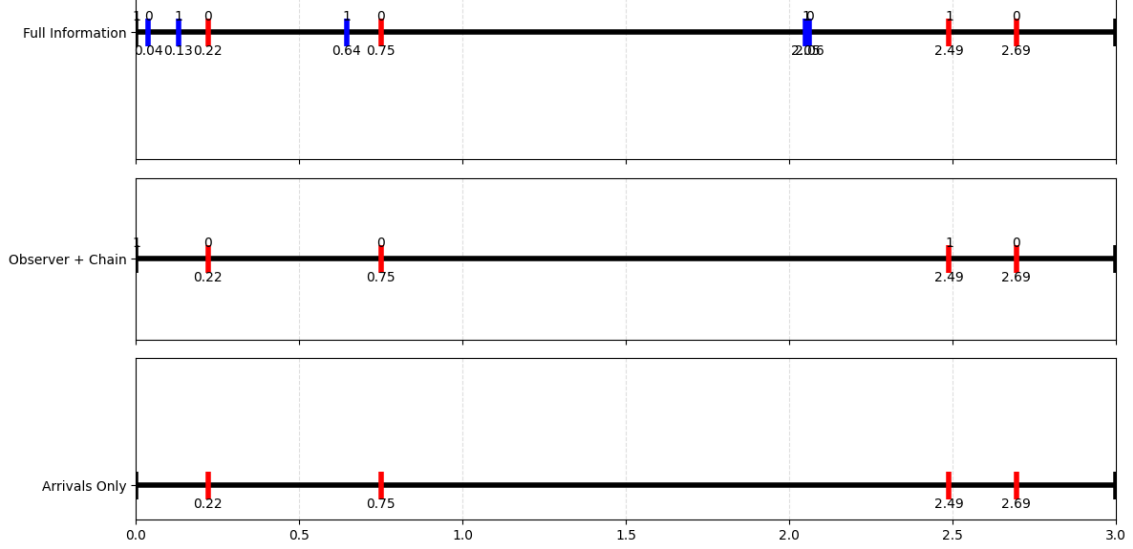


Fig. 3.1. Run of the MAP given by the previous parameters of Example 2.

The inter-arrival times of the MAP (technically, the MAP's counting process) have been studied extensively, and their distributions are well-known. They follow what are known as phase-type distributions (PH).

Definition 18: Let X be a non-negative random variable, $m \in \mathbb{N}$, $\alpha \in \mathbb{R}^m$ a probability vector, and $T \in \mathbb{R}^{m \times m}$ an invertible matrix with negative diagonal elements, non-negative off-diagonal elements, and non-positive row sums. Then, X is said to be *phase-type distributed* with parameters (α, T) (or, following a “PH- (α, T) ” distribution) if its distribution function F is given by the formula

$$F(x) = 1 - \alpha e^{xT} \mathbf{1}, \quad \forall x \in \mathbb{R}.$$

The proofs for the derivations of the distribution functions of MAP inter-event times may be consulted in [4] (page 12).

Proposition 7: Let $\mathcal{M}$ be a MAP, represented by the parameters (α_0, D_0, D_1) , and let $\{T_n\}_{n \in \mathbb{N}}$ be its sequence of inter-event times. Then, for all $n \in \mathbb{N}$, T_n follows a $PH(\alpha_{n-1}, D_0)$ distribution, where $\alpha_{n-1} = \alpha_0(P^*)^{n-1}$.

From Proposition 7, formulas for the moments, density functions, and many other probabilistic objects can be immediately derived. They may be consulted in standard textbooks about MAPs [21, Chapter 1].

4. THE CLUSTERED-STATE MARKOVIAN ARRIVAL PROCESS

In this chapter, we present the RD CS-MAPs. The literature on this topic is scarce, so our results will be based on those from previous Master's theses on this topic (see [24], [3] and [4]).

4.1. Definition

The Clustered-States MAP (CS-MAP) is a type of point process that models lifeline datasets where there is more than one kind of event, in that events carry labels. This is different from the Poisson process, whose unique events, the *arrivals*, ended the process; and also from the MAPs, which couldn't distinguish between different types of events. In the Clustered-States MAP, we have the same setup we had with the MAP: a continuous-time Markov chain and a counting process which registered transitions with arrivals; with the key difference that only the clusters to which the states belong are considered, rather than the states themselves. By understanding the MAP, the formal definition for the CS-MAP follows naturally.

Definition 19: Let $\{(I(t), N(t)) | t \geq 0\}$ be a MAP with state space S , and let $\sigma: S \rightarrow S'$ be a surjective function. Then, the bivariate stochastic process $\{(\sigma(I(t)), N(t)) | t \geq 0\}$ is known as a **Clustered-States Markovian Arrival Process (CS-MAP)**, with state space S' . Moreover, if $|S| = m$, we say that the CS-MAP is a CS-MAP $_m$.

We should look at some examples.

Example 4: Every Poisson process can be seen both as a MAP $_1$ and a CS-MAP $_1$ because there is only one function from a singleton set onto another singleton set, and it is surjective. Next, considering the MAP $_2$, we can only have two different CS-MAP $_2$ s: the one with σ mapping $\{1, 2\}$ onto $\{1, 2\}$ bijectively, or the one mapping $\{1, 2\}$ onto $\{1\}$ surjectively. Neither CS-MAP is interesting, as the first is equivalent to the MAP $_2$, and the second one is trivial in that we don't label events in a novel way (however, it is not a MAP). In general, we do not want CS-MAPs to only have a single cluster-state, nor do we want a bijective clustering since this would just produce a MAP. These are all considered trivial examples, which means that the first interesting CS-MAPs are at least 3-dimensional.

Example 5: Consider the MAP $_3$ given by the parameters:

$$\alpha_0 = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}, \quad D_0 = \begin{bmatrix} -2.5 & 2 & 1 \\ 3 & -12 & 1 \\ 1 & 2 & -8 \end{bmatrix}, \quad D_1 = \begin{bmatrix} 0 & 0.5 & 0 \\ 7 & 0 & 1 \\ 1 & 3 & 1 \end{bmatrix}.$$

We will choose two distinct CS-MAP₃s. The first one comes from the clusters {1, 2}, {3}, and the second one comes from the clusters {1, 3}, {2}. Since the matrices are not symmetric, each original state has transition rates distinct from those from the other states, hence these two clusterings will produce two completely different stochastic processes.

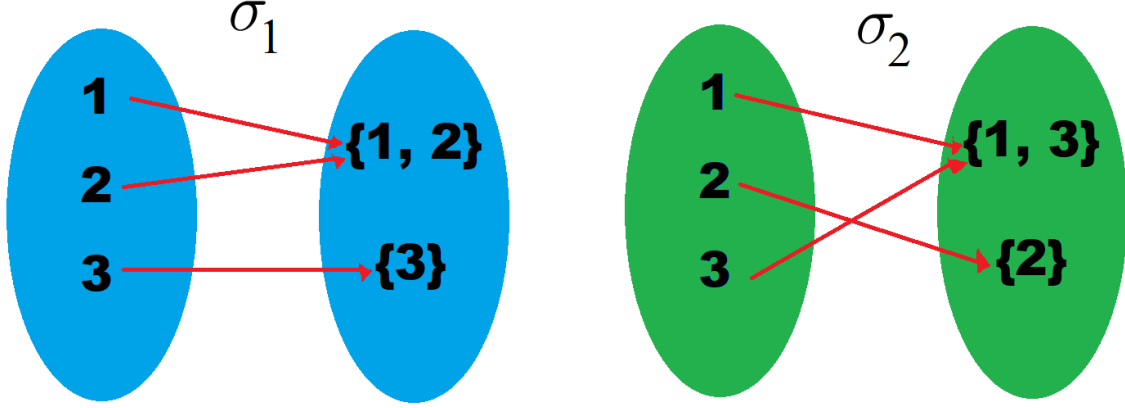


Fig. 4.1. CS-MAP clustering functions for the given MAP₃.

Example 6: Consider a MAP _{m} and any surjective function onto any 2-element set. In this setup, we want one of the clusters to be an absorbing cluster-state, and the other to be a transient cluster-state. CS-MAPs formed by such maps and surjective functions shall be denoted as *Recurrence-Death CS-MAPs* (RD CS-MAPs), because the cluster-states set can be replaced by the set $\{R, D\}$, the labels for the *recurrence* event and the terminal *death* event. An interesting fact about RD CS-MAPs is that each RD CS-MAP where more than 1 of the original states has been assigned to the cluster-state D is equivalent to an RD CS-MAP with exactly 1 of the original states assigned to the cluster-state D , and all others to the cluster-state R . Lastly, taking this into account, we may assume without loss of generality that the last original state m is the unique state such that $\sigma(m) = D$ (i.e. the unique *death* state).

This last example is very important because it is the kind of point process that we need for the cancer datasets. Not only do we need a CS-MAP so that we can label recurrence and death events correctly, but we also need the Recurrence-Death version of the CS-MAP so that the death event is modeled as a terminal event. Therefore, the rest of this thesis's contents will be focused on RD CS-MAPs.

4.2. The Recurrence-Death CS-MAP

In the last section we introduced the RD CS-MAP as a particular case of the CS-MAP. We will summarize the previous discussion within a formal definition.

Definition 20: Let $\{(I(t), N(t)) | t \geq 0\}$ be a MAP _{m} and $\sigma: \{1, \dots, m\} \rightarrow \{R, D\}$ the surjective function defined by $\sigma(m) = D$ and $\sigma(i) = R$ for all $1 \leq i \leq m-1$. Moreover, for

the Markov chain $\{I(t) | t \geq 0\}$, let the state m be absorbent and all other states $1, \dots, m-1$ be transient². Then, the bivariate stochastic process $\{(\sigma(I(t)), N(t)) | t \geq 0\}$ is known as a *Recurrence-Death CS-MAP (RD CS-MAP)*.

The RD CS-MAP is just a CS-MAP with the condition that it must only have two labels, and one of them is absorbent and the other one is not. Without loss of generality, the death cluster state is composed of a single original state, which allows for streamlining the theory.

In general CS-MAPs, we may always define a discrete random variable, which is the minimum number of occurrences of a certain type of event before observing an occurrence of another type of event. However, this variable is much more important for RD CS-MAPs, since its value tells us how many recurrences we have to observe before the death event, which ends the simulation. Not only that, but it also governs both the visited states sequence and the inter-event times' distributions. In the context of RD CS-MAPs, this is the minimum number of recurrences until death.

Definition 21: Consider an RD CS-MAP with original Markov chain $\{I(t) | t \geq 0\}$ and sequence of original state transition times $\{t_n\}_{n=1}^\infty$ ³. We define the (*minimum*) *number of recurrences until death* of the RD CS-MAP as

$$r = \min\{n \in \mathbb{N} | I(t_{n+1}) = m\} = \min\{n \in \mathbb{N} | \sigma(I(t_{n+1})) = D\}.$$

From here onwards we introduce some important notation: I_R is the identity matrix with a 0 on its last diagonal element, and I_D the zero matrix with a 1 on its last diagonal element. Note that the relation $I_R + I_D = I$ holds.

The distribution of the random variable r had been completely determined in previous work.

Theorem 2: Consider an RD CS-MAP with parameter set (α_0, D_0, D_1) , and let r be its number of recurrences until death. Then:

$$\begin{aligned}\mathbb{P}(r > n) &= \alpha_0(P^* I_R)^{n+1} \mathbb{1} \\ \mathbb{P}(r = n) &= \alpha_0(P^* I_R)^n P^* I_D \mathbb{1}\end{aligned}$$

The expected value of r had also been calculated in previous work. Concise proofs of the survival function, mass function and expected value of r can be found in [4, pages 17-18].

²It follows that, for the Markov chain $\{\sigma(I(t)) | t \geq 0\}$, the state D is absorbent and the state R is transient.

³Thus, $\{I(t_n)\}_{n=1}^\infty$ is the sequence of visited original states, and $\{\sigma(I(t_n))\}_{n=1}^\infty$ is the sequence of visited cluster states.

Our new result is the calculation of higher moments of r , which allows in particular for the calculation of $\mathbb{V}(r)$, and other quantities. Furthermore, we determined that r 's distribution satisfies the “moments problem” of Probability Theory, which states that r 's distribution can also be determined by its infinite sequence of moments.

Theorem 3: Let r be the number of recurrences until death of an RD CS-MAP with parameter set (α_0, D_0, D_1) . Then, for all $k \in \mathbb{N}$:

$$\mathbb{E}(r^k) = \alpha_0 \left(\sum_{j=1}^k j! \left\{ \begin{matrix} k \\ j \end{matrix} \right\} (P^* I_R (I - P^* I_R)^{-1})^j \right) \mathbb{1}, \quad (4.1)$$

where $\left\{ \begin{matrix} k \\ j \end{matrix} \right\}$ is the “ k over j ” Stirling number of the second kind.

In particular:

$$\mathbb{E}(r) = \alpha_0 P^* I_R (I - P^* I_R)^{-1} \mathbb{1}.$$

Moreover, let X_1 and X_2 be random variables such that for all $k \in \mathbb{N}$: $\mathbb{E}(X_1^k)$ and $\mathbb{E}(X_2^k)$ can be given by Equation (4.1). Then X_1 and X_2 are equal in distribution (i.e. the distribution functions of X_1 and X_2 are equal).

From formula (4.1), quantities such as $\mathbb{V}(r)$ or the skewness coefficient of r may be immediately calculated analytically..

Next, we exhibit the results known about the inter-event times. The key difference that RD CS-MAP inter-event times have with CS-MAP inter-event times is that RD CS-MAP inter-event times are only meaningful subject to r 's values. For example, if for a given RD CS-MAP we have that $r = 3$, we understand that $T_4|_{r=3}$ is a death time, so the patient passed away at time $T_4|_{r=3}$, and $T_k|_{r=3} = 0$ for all $k \geq 5$.

One of the most important differences between MAPs and CS-MAPs is that CS-MAP inter-event times are not phase-type distributed. However, explicit formulas for their distribution functions for all different possible scenarios have been determined.

Theorem 4: Let $n \in \mathbb{N}$, let r be the number of recurrences until death of an RD CS-MAP, and let $T_1, \dots, T_n$ be the first n inter-event times of the same RD CS-MAP. We have:

$$\begin{aligned} F_{T_1, \dots, T_n | r \geq l}(t_1, \dots, t_n) &= (-1)^n \frac{\alpha_0 \left(\prod_{i=1}^n e^{t_i D_0} P^* I_R \right) (P^* I_R)^{l-n} \mathbb{1}}{\alpha_0 (P^* I_R)^l \mathbb{1}}, \quad \forall l \geq n \\ F_{T_1, \dots, T_n | r \geq n-1}(t_1, \dots, t_n) &= \frac{\alpha_0 \left(\prod_{i=1}^{n-1} (I - e^{t_i D_0}) P^* I_R \right) (I - e^{t_n D_0}) \mathbb{1}}{\alpha_0 (P^* I_R)^{n-1} \mathbb{1}} \\ F_{T_1, \dots, T_n | r = n-1}(t_1, \dots, t_n) &= \frac{\alpha_0 \left(\prod_{i=1}^{n-1} (I - e^{t_i D_0}) P^* I_R \right) (I - e^{t_n D_0}) P^* I_D \mathbb{1}}{\alpha_0 (P^* I_R)^{n-1} P^* I_D \mathbb{1}} \end{aligned}$$

In particular:

$$\begin{aligned} F_{T_n | r \geq l}(t) &= -\frac{\alpha_0(P^* I_R)^{n-1} e^{tD_0} (P^* I_R)^{l-(n-1)} \mathbb{1}}{\alpha_0(P^* I_R)^l \mathbb{1}}, \quad \forall l \geq n \\ F_{T_n | r \geq n-1}(t) &= -\frac{\alpha_0(P^* I_R)^{n-1} e^{tD_0} \mathbb{1}}{\alpha_0(P^* I_R)^{n-1} \mathbb{1}} \\ F_{T_n | r = n-1}(t) &= -\frac{\alpha_0(P^* I_R)^{n-1} e^{tD_0} P^* I_D \mathbb{1}}{\alpha_0(P^* I_R)^{n-1} P^* I_D \mathbb{1}} \end{aligned}$$

A detailed derivation of these formulas may be found in [3].

With formulas for the conditional distribution functions of the inter-event times, we may immediately obtain formulas for the density functions.

Proposition 8: Let $n \in \mathbb{N}$, let r be the number of recurrences until death of an RD CS-MAP, and let $T_1, \dots, T_n$ be the first n inter-event times of the same RD CS-MAP. Then:

$$\begin{aligned} f_{T_1, \dots, T_n | r \geq l}(t_1, \dots, t_n) &= (-1)^n \frac{\alpha_0 \left(\prod_{i=1}^n D_0 e^{t_i D_0} P^* I_R \right) (P^* I_R)^{l-n} \mathbb{1}}{\alpha_0(P^* I_R)^l \mathbb{1}}, \quad \forall l \geq n \\ f_{T_1, \dots, T_n | r \geq n-1}(t_1, \dots, t_n) &= \frac{\alpha_0 \left(\prod_{i=1}^{n-1} e^{t_i D_0} D_1 I_R \right) e^{t_n D_0} D_1 \mathbb{1}}{\alpha_0(P^* I_R)^{n-1} \mathbb{1}}, \\ f_{T_1, \dots, T_n | r = n-1}(t_1, \dots, t_n) &= \frac{\alpha_0 \left(\prod_{i=1}^{n-1} e^{t_i D_0} D_1 I_R \right) e^{t_n D_0} D_1 I_D \mathbb{1}}{\alpha_0(P^* I_R)^{n-1} P^* I_D \mathbb{1}}. \end{aligned}$$

In particular,

$$\begin{aligned} f_{T_n | r \geq l}(t) &= -\frac{\alpha_0(P^* I_R)^{n-1} D_0 e^{tD_0} (P^* I_R)^{l-(n-1)} \mathbb{1}}{\alpha_0(P^* I_R)^l \mathbb{1}}, \quad \forall l \geq n \\ f_{T_n | r \geq n-1}(t) &= -\frac{\alpha_0(P^* I_R)^{n-1} D_0 e^{tD_0} \mathbb{1}}{\alpha_0(P^* I_R)^{n-1} \mathbb{1}}, \\ f_{T_n | r = n-1}(t) &= -\frac{\alpha_0(P^* I_R)^{n-1} D_0 e^{tD_0} P^* I_D \mathbb{1}}{\alpha_0(P^* I_R)^{n-1} P^* I_D \mathbb{1}}. \end{aligned}$$

Going further, with formulas for the conditional distribution and density functions of the inter-event times, it is plausible to find formulas for the conditional moments of the inter-event times.

Proposition 9: Let $n \in \mathbb{N}$, let r be the number of recurrences until death of an RD CS-MAP, and let $T_1, \dots, T_n$ be the first n inter-event times of the same RD CS-MAP. For all $k \in \mathbb{N}$ we have:

$$\begin{aligned} \mathbb{E}(T_n^k | r \geq l) &= (-1)^{k+1} k! \frac{\alpha_0(P^* I_R)^{n-1} D_0^{-k} (P^* I_R)^{l-(n-1)} \mathbb{1}}{\alpha_0(P^* I_R)^l \mathbb{1}}, \quad \forall k \geq 1, \quad l \geq n-1 \\ \mathbb{E}(T_n^k | r \geq n-1) &= (-1)^{k+1} k! \frac{\alpha_0(P^* I_R)^{n-1} D_0^{-k} \mathbb{1}}{\alpha_0(P^* I_R)^{n-1} \mathbb{1}}, \quad \forall k \geq 1 \\ \mathbb{E}(T_n^k | r = n-1) &= (-1)^{k+1} k! \frac{\alpha_0(P^* I_R)^{n-1} D_0^{-k} P^* I_D \mathbb{1}}{\alpha_0(P^* I_R)^{n-1} P^* I_D \mathbb{1}}, \quad \forall k \geq 1 \end{aligned}$$

Lastly, we can also calculate conditional expected values of products of times. The main use of these formulas is for the calculation of conditional inter-event time correlations.

Proposition 10: Let $n \in \mathbb{N}$, let r be the number of recurrences until death of an RD CS-MAP, and let $T_1, \dots, T_n$ be the first n inter-event times of the same RD CS-MAP. For all $k \in \mathbb{N}$ we have:

$$\begin{aligned}\mathbb{E}(T_n T_p | r \geq l) &= \frac{\alpha_0 (P^* I_R)^{n-1} D_0^{-1} (P^* I_R)^{p-n} D_0^{-1} (P^* I_R)^{l-(p-1)} \mathbb{1}}{\alpha_0 (P^* I_R)^l \mathbb{1}}, & \forall p > n \geq 1, l \geq p-1 \\ \mathbb{E}(T_n T_p | r \geq p-1) &= \frac{\alpha_0 (P^* I_R)^{n-1} D_0^{-1} (P^* I_R)^{p-n} D_0^{-1} \mathbb{1}}{\alpha_0 (P^* I_R)^{p-1} \mathbb{1}}, & \forall p > n \geq 1 \\ \mathbb{E}(T_n T_p | r = p-1) &= \frac{\alpha_0 (P^* I_R)^{n-1} D_0^{-1} (P^* I_R)^{p-n} D_0^{-1} P^* I_D \mathbb{1}}{\alpha_0 (P^* I_R)^{p-1} P^* I_D \mathbb{1}}, & \forall p > n \geq 1\end{aligned}$$

With these last quantities, we may analytically calculate the conditional inter-event time correlations. The conditional inter-event time correlations are especially important in this thesis, because these will be used for statistical inference in a future section, a new idea we believe has never been reported before outside of CS-MAP literature. Derivations and expositions of all of the previous formulas may all be found in [4]. A derivation of the Laplace-Stieltjes function is also derived in [24], not featured in this thesis.

4.3. The effect of inter-event time correlations on $\mathbb{E}(r)$

Our contributions to this area are focused around discerning whether the number of recurrences until death of RD CS-MAPs behave differently when the inter-event times are highly correlated compared to when they are not. The mean, standard deviation and 95th percentile of r were chosen as the representatives of r 's distribution because from Theorem 3 we know that r 's distribution is characterized uniquely by its moments, and the 95th percentile is useful for analyzing the tail. Moreover, the first three conditional inter-event time means were fixed as a control variable in the experiment because it is not known whether these can have an effect on the conditional inter-event time correlations or r 's statistics.

First, a study on the behaviour of the quantity $\mathbb{E}(r)$ was conducted. The following steps were performed:

1. Search for MAP parameters (α_0, D_0, D_1) until a trio such that the first three inter-event time conditional correlations $\rho(T_1, T_2 | r \geq 1)$, $\rho(T_1, T_3 | r \geq 2)$, $\rho(T_2, T_3 | r \geq 2)$ were very close to 0.
2. Calculate the first three inter-event time conditional means for the previous parameter set: $\mathbb{E}(T_1 | r \geq 0)$, $\mathbb{E}(T_2 | r \geq 1)$, $\mathbb{E}(T_3 | r \geq 2)$. These three values are memorized for the rest of the experiment, call them μ_1, μ_2, μ_3 respectively.

3. Search for MAP parameters (α_0, D_0, D_1) with their first three inter-event time means being close to μ_1, μ_2, μ_3 respectively. When such parameters are found, their first three inter-event time conditional correlations are calculated, and put into a “low correlations group” if they were all very close to 0, and put into a “high correlations group” if the sum of absolute values of the correlations was higher than a certain threshold. If the correlations were not classified as either low or high, the parameter set was discarded altogether.
4. For parameter sets with means close to the fixed values, and either sufficiently low or high correlations, their associated $\mathbb{E}(r)$ value was calculated.

In more concrete terms, in the code used, the parameters for the experiment were the following:

- **ORDER:** Dimensions of the MAP parameters.
- **GAMMA_SHAPE, GAMMA_SCALE:** Parameters of the Gamma distribution from which the off-diagonal elements of the D_0 matrix and elements of the D_1 matrix were sampled.
- **CORR_THRESHOLDS:** Lower bounds for classifying each of the three correlation values as “high”. We considered a trio of correlations as “high” if the sum of the absolute values surpassed the sum of the three bounds. This variable is a list because initially we worked with individual correlation values, even though in the end we considered all correlations and all bounds jointly.
- **FIRST_CORR_TOL, SECOND_CORR_TOL:** Tolerance values for the absolute errors of “low” correlations. The first parameter is used for the parameter set draws from before fixing the means, and the second parameter is used for the parameter set draws from after fixing the means.
- **MEAN_TOL:** Tolerance values for the relative errors of the drawn parameter set means from the fixed means. For instance, a value of “2” means that values are not allowed to be twice as small or large as the original value. Relative error was used because inter-event time means may be arbitrarily large.
- **SAMPLE_SIZE:** Sizes of the low and high correlations parameter set lists. Both lists contained the same amount of parameter sets.
- **MAX_ITERS:** Maximum number of parameter sets the computer would draw. This is just a safety measure.
- **MASTER_SEED:** Random Number Generator seed which ensured the reproducibility of the experiments.

A simulation with some fixed experiment parameters was done, those being:

ORDER = 3	SECOND_CORR_TOL = $1e-3$
GAMMA_SHAPE = 1	MEAN_TOL = 2
GAMMA_RATE = 0.5	SAMPLE_SIZE = 500
CORR_THRESHOLDS = [0.2, 0.2, 0.2]	MAX_ITERS = $6 \cdot 10^8$
FIRST_CORR_TOL = $1e-3$	MASTER_SEED = 42

and the drawn means were

$$\mu_1 = 6.7120$$

$$\mu_2 = 5.8378$$

$$\mu_3 = 5.8453$$

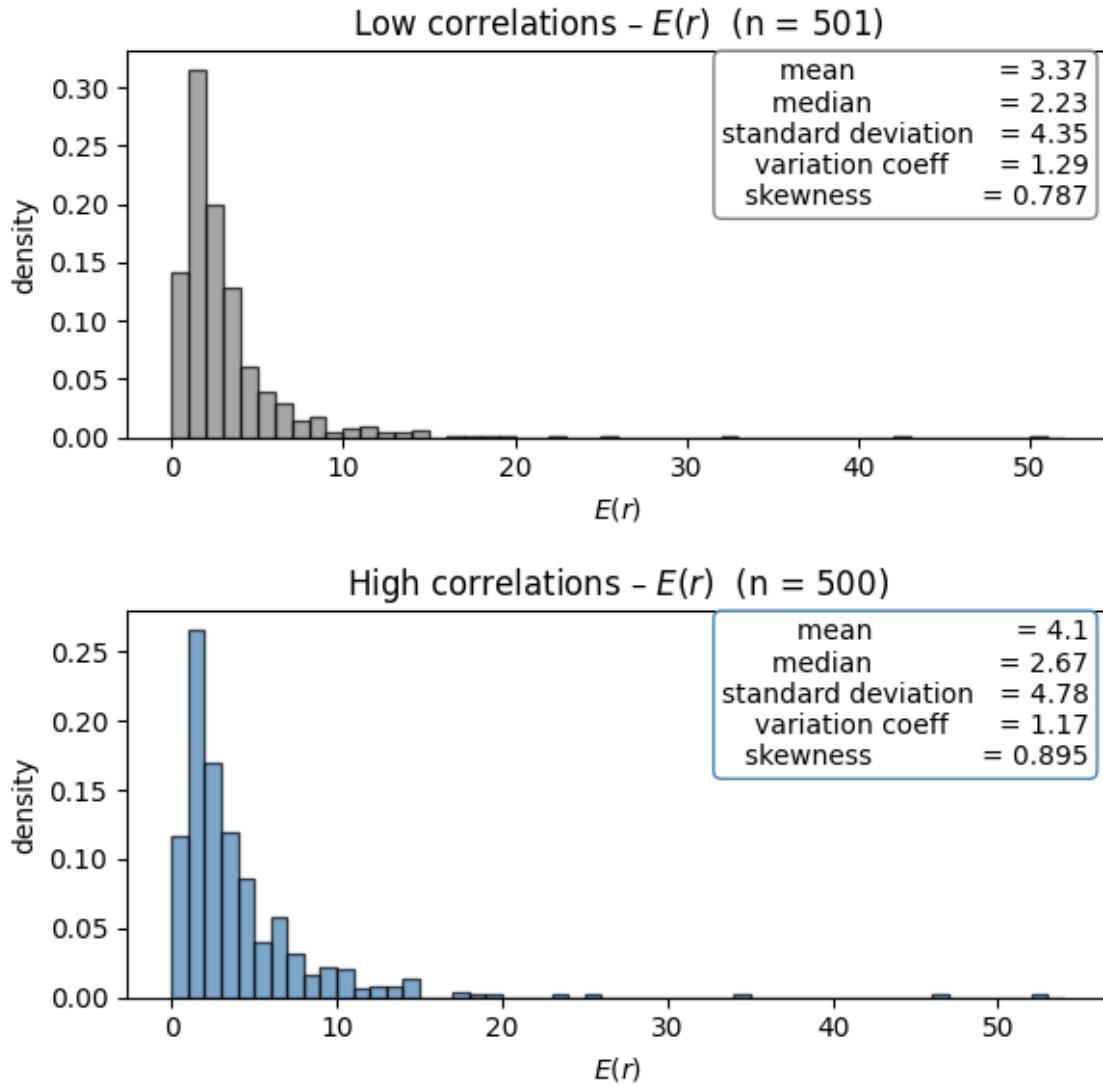


Fig. 4.2. Histograms of the values of $\mathbb{E}(r)$, the mean of r , for different RD CS-MAP parameter sets.

Execution time was approximately 20 hours. The statistics related to the histograms' empirical distributions were also calculated, and have been displayed in both images. The obtained conclusion was that there was not enough statistical evidence to support the fact that inter-event time conditional correlations had an effect on the distribution of the expected value of the number of recurrences until death. This means that the experiment would have to be repeated with stricter bounds for the high correlations, which could end up multiplying the execution time. An alternative is to modify the Gamma distribution parameters (or choosing a different distribution, or even an entirely new sampling method) if these are discovered to have an effect on the correlations, but this is also unknown and would take a lot of development time.

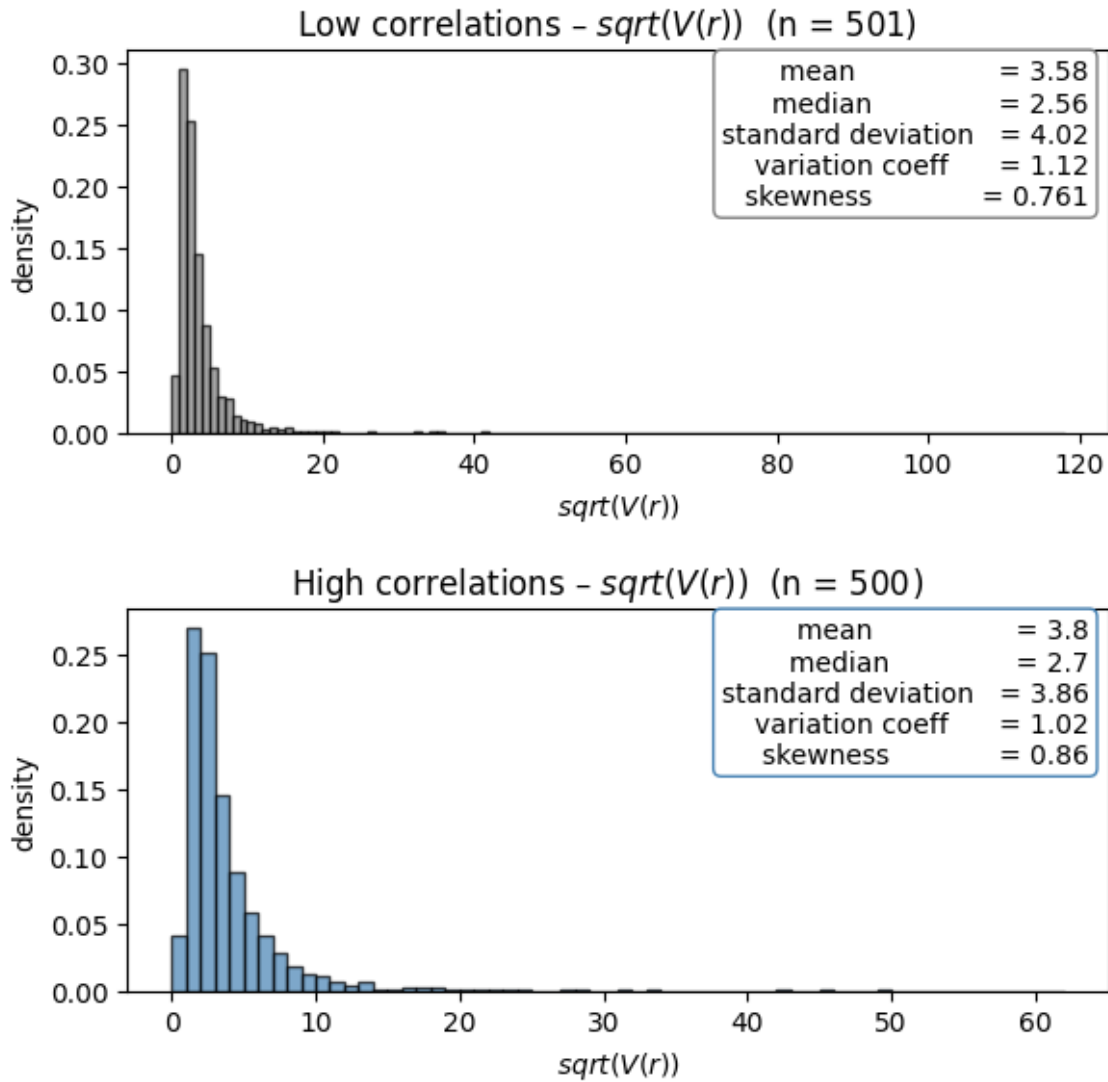


Fig. 4.3. Histograms of the values of $\sqrt{V(r)}$, the standard deviation of r , for different RD CS-MAP parameter sets.

Using the previously determined formula for the moments of r , $\mathbb{E}(r^k)$, an analogous study was conducted for $\sqrt{V(r)}$, the standard deviation of r , and the result was also inconclusive, just as before.

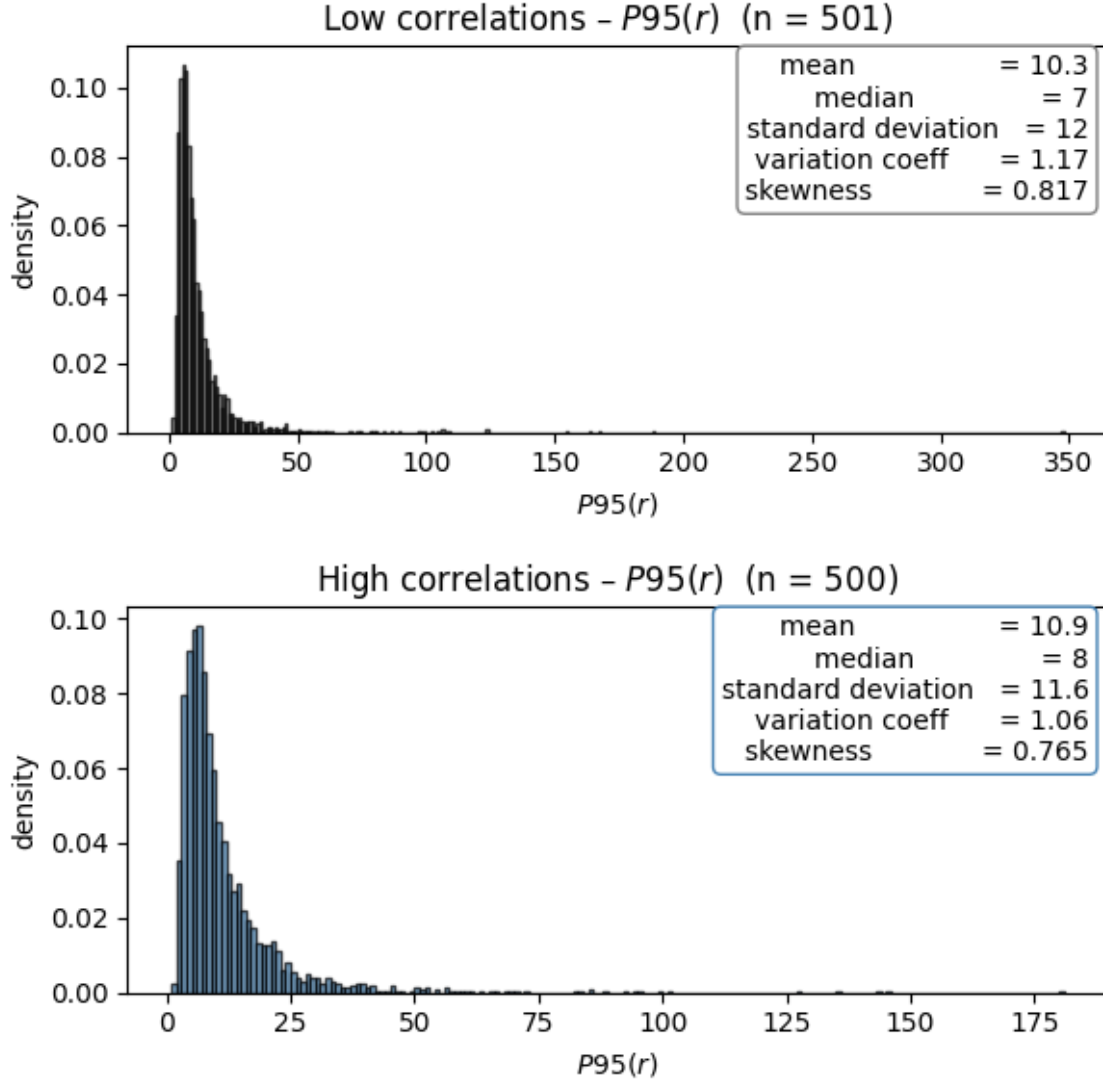


Fig. 4.4. Histograms of the values of the 95th percentile of r for different RD CS-MAP parameter sets.

In the search for differences, one last comparison was tried, this time with the 95th percentile of r . This expression has no explicit formula, but this was not a problem, since it can be easily calculated via a numerical bisection method of the distribution function, for which we do have a formula. However, once again, no significant changes in the distributions were observed.

In retrospect, we believe that stricter values of the CORR_THRESHOLDS should have been enforced, at the expense of more execution time, because absolute values of 0.2 for high correlations might be too low. However, execution time might have become unbearably high, so only a complete overhaul of the sampling method or a much bigger investment in computer resources could have helped us to carry out a more conclusive experiment.

4.4. The effect of inter-event time correlations on conditional death times

After the previous experiment, another related but different experiment was devised. The aim was to discern whether the inter-event time correlations have a meaningful effect on the time until death, conditioned on the first time being relatively large.

In more technical terms, the time until death T_D is a random variable defined as $T_D = \sum_{i=1}^{\infty} T_i$. It must be noted that all terms in the sum except for finitely many are 0, so T_D is just the finite sum of all inter-event recurrence times and the inter-event death time. Our goal was to study $\mathbb{E}(T_D | T_1 > t)$. The reason for this choice is the following: if T_1 is large and T_1, T_2 and T_3 are all highly correlated, then we should also expect T_2 and T_3 to be large, and in the end result, $\mathbb{E}(T_D | T_1 > t)$ should be very large; while if T_1, T_2 and T_3 are not correlated, then we shouldn't expect $\mathbb{E}(T_D | T_1 > t)$ to obtain large values frequently. In short: this experiment would reveal that RD CS-MAP conditional inter-event time correlations have an effect on death times - high correlations would extend the death times.

Let $t > 0$ be the desired lower bound on T_1 , tuned by the user. The following steps were performed:

1. Search for MAP parameters (α_0, D_0, D_1) with their associated $\mathbb{E}(r)$ value close to a fixed one. The previous experiment helped us to decide that 2.15 was an optimal value, since this value is close to the medians and modes of the histograms from the previous experiments, meaning that we would find more parameters sets overall. When such parameter sets are found, their first three inter-event time conditional correlations are calculated, and put into a “low correlations group” if they were all very close to 0, and put into a “high correlations group” if the sum of their absolute values was higher than a certain threshold. If the correlations were not classified as either low or high, the parameter set was discarded altogether.
2. For parameter sets with means close to the fixed values, and either sufficiently low or high correlations, their associated $\mathbb{E}(T_D | T_1 > t)$ was estimated. Since explicit formulas for this quantity are unknown, MonteCarlo simulations were carried out to obtain empirical $\overline{T_D}|_{T_1 > t}$ values, and these were used as estimates for the theoretical values.

In more concrete terms, in the code used, the parameters for the experiment were the following:

- ORDER: Dimensions of the MAP parameters.
- GAMMA_SHAPE, GAMMA_SCALE: Parameters of the Gamma distribution from which the off-diagonal elements of the D_0 matrix and elements of the D_1 matrix were sampled.

- **CORR_THRESHOLDS**: Lower bounds for classifying each of the three correlation values as “high”.
- **CORR_TOL**: Tolerance value for the absolute error of the “low” correlations.
- **FIXED_MEAN**: The fixed value of $\mathbb{E}(r)$.
- **MEAN_TOL**: Tolerance values for the relative errors of the drawn parameter set means from the fixed $\mathbb{E}(r)$ value. Absolute error was used because fixing the mean appropriately was more important in this experiment.
- **SAMPLE_SIZE**: Sizes of the low and high correlations parameter set lists. Both lists contained the same amount of parameter sets.
- **MAX_ITERS**: Maximum number of parameter sets the computer would draw. Tuning it for this experiment is harder, because MonteCarlo simulation results can be empty if no simulation satisfied the imposed “ $T_1 > t$ ” condition.
- **K_MAX**: Maximum number of recurrences per RD CS-MAP run. This is just a safety parameter. Ideally, if storage was not a problem, we would make it as large as possible.
- **N_PATHS**: Maximum number of RD CS-MAP runs for averaging. Tuning this parameter turned out to be very difficult but also very important, since it dictated the quality of our MonteCarlo estimates for the theoretical quantities. If it had been too low, there would not have been enough RD CS-MAP runs satisfying $T_1 > t$ for obtaining a decent estimate, if any at all; while if it had been too high, the experiment could have taken an unexpectedly long amount of execution time.
- **COND_TIME**: The lower bound on T_1 , $t > 0$. This is the other critical parameter: if it had been too low, then $\mathbb{E}(T_D | T_1 > t) \approx \mathbb{E}(T_D)$, and it would not have been possible to appreciate the potential effects of inter-event time correlations; while if it had been too high, MonteCarlo estimates could have decreased drastically in quality from sample size reductions, which would cause the execution time of the experiment to greatly increase.
- **MASTER_SEED**: Random Number Generator seed which ensured the reproducibility of the experiments.

Two simulations with some fixed experiment parameters were done, which were:

ORDER = 3	FIXED_MEAN = 2.15
GAMMA_SHAPE = 1	MEAN_TOL = 0.15
GAMMA_RATE = 0.5	SAMPLE_SIZE = 500
CORR_THRESHOLDS = [0.2, 0.2, 0.2]	MAX_ITERS = $9 \cdot 10^{**}8$
CORR_TOL = $1e - 3$	COND_TIME $\in \{0.1, 0.5\}$
KMAX = 200	N_PATHS $\in \{10^{**}5, 4 \cdot 10^{**}5\}$
MASTER_SEED = 42	

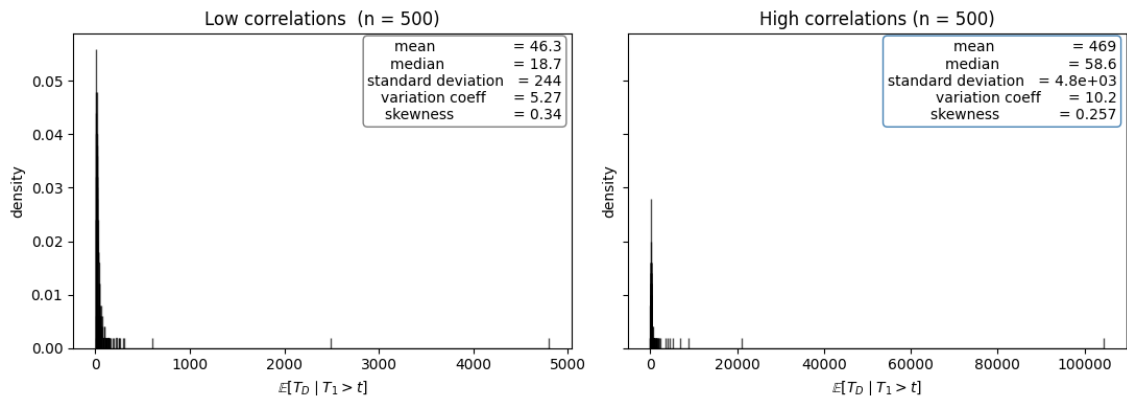


Fig. 4.5. Histograms of the values of $\mathbb{E}(T_D | T_1 > t)$ for different RD CS-MAP parameter sets and threshold $t = 0.1$.

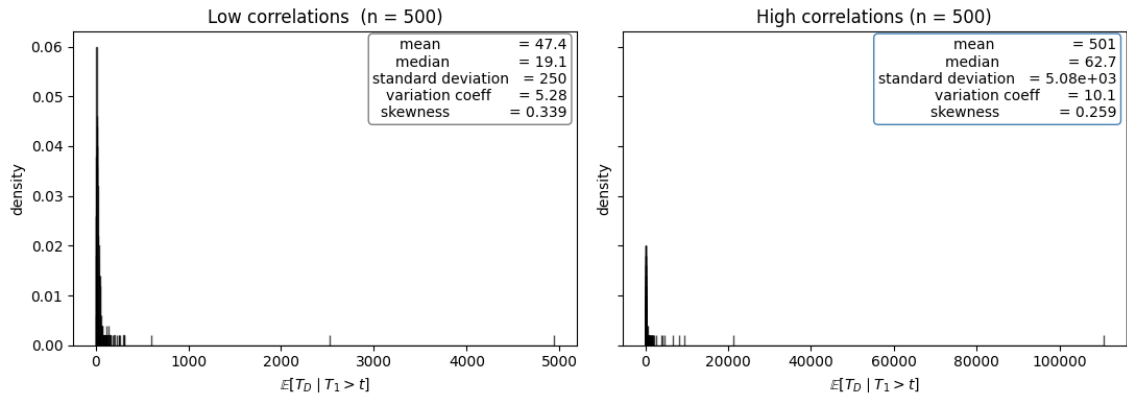


Fig. 4.6. Histograms of the values of $\mathbb{E}(T_D | T_1 > t)$ for different RD CS-MAP parameter sets and threshold $t = 0.5$.

Execution time was roughly 20 hours and 28 hours for each. We found out that correlations did have an effect on the distributions of the death times. The effects can be appreciated in the histogram statistics, even though they look similar, the high correlations histogram has a much larger standard deviation, so its values are larger.

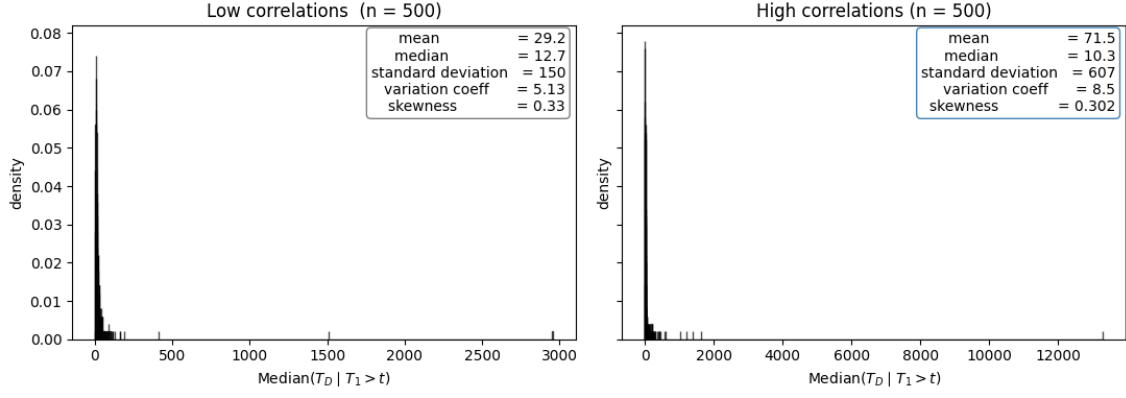


Fig. 4.7. Histograms of the values of $\text{Median}(T_D | T_1 > t)$ for different RD CS-MAP parameter sets and threshold $t = 0.1$.

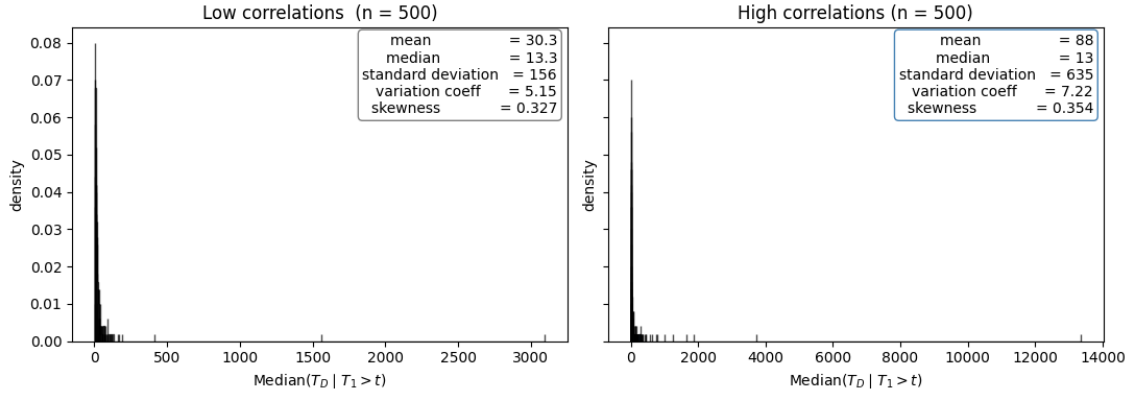


Fig. 4.8. Histograms of the values of $\text{Median}(T_D | T_1 > t)$ for different RD CS-MAP parameter sets and threshold $t = 0.5$.

We then repeated this experiment for $\text{Median}(T_D | T_1 > t)$, the median death time conditioned upon the fact that T_1 is larger than the threshold t . Even though the median is more robust than the mean, the high correlations histogram still features a much larger standard deviation, suggesting that $\text{Median}(T_D | T_1 > t)$ sometimes reaches large values. This further reinforces our conclusion that inter-event time correlations have an effect on the conditional death time's distribution.

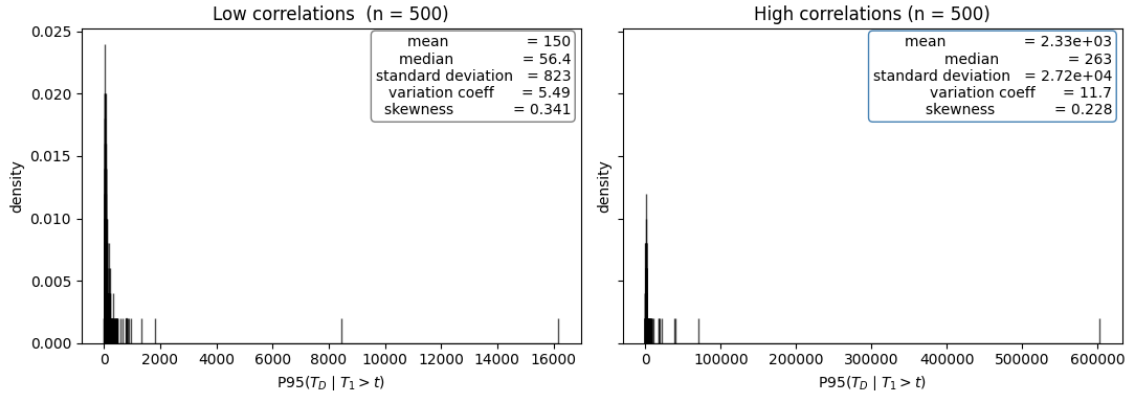


Fig. 4.9. Histograms of the values of $(\text{Percentile } 95)(T_D | T_1 > t)$ for different RD CS-MAP parameter sets and threshold $t = 0.1$.

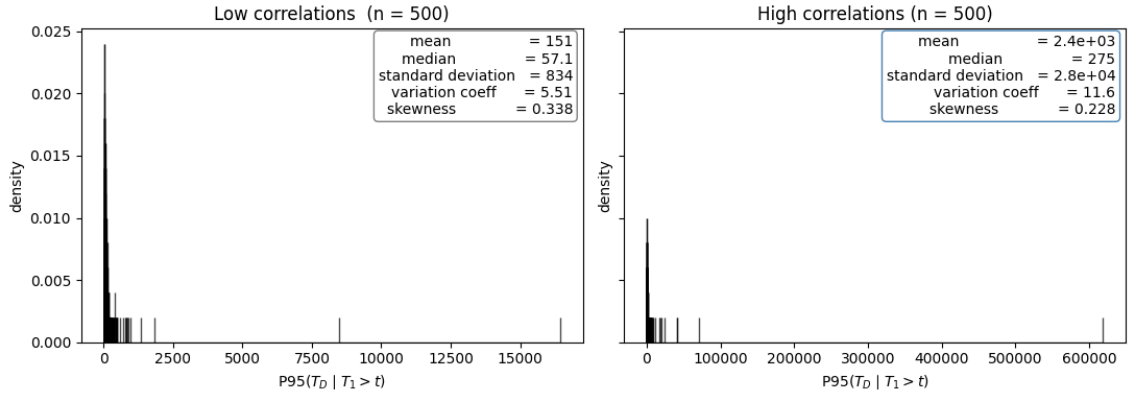


Fig. 4.10. Histograms of the values of $(\text{Percentile } 95)(T_D | T_1 > t)$ for different RD CS-MAP parameter sets and threshold $t = 0.5$.

Lastly, the same experiment was carried out for $(\text{Percentile } 95)(T_D | T_1 > t)$, the 95th percentile of the death time conditioned on the fact that T_1 is larger than a certain threshold t . The obtained conclusions were consistent with those from the mean and median: inter-event time correlations have an effect on the conditional distribution of the death time.

5. STATISTICAL INFERENCE FOR THE CS-MAP

In this last section, we review and explore some inference methods for RD CS-MAPs applied to lifeline datasets. First, we discuss the basis and the procedures of the experiments we carried out. Then, we present the results for three different datasets: simulated samples without censorship, simulated samples with censorship, and real datasets.

5.1. The loglikelihood function

Since CS-MAPs are relatively novel, results about statistical inference for CS-MAPs are scarce. Work in this topic began with the derivation of the formula for the loglikelihood function of an i.i.d. sample of RD CS-MAP conditional inter-event times.

Theorem 5: (Loglikelihood function of an RD CS-MAP sample)

Let $m \in \mathbb{N}$, and consider a RD CS-MAP _{m} with Markov chain $I(t)$, and represented by the parameter set (α_0, D_0, D_1) and clustering function σ . Also, let $n, p \in \mathbb{N}$ and let $\Psi = (\psi_1, \dots, \psi_p)$ be a p -sized i.i.d. sample of the recurrence and death inter-event times of the previous RD CS-MAP; that is, for every element ψ_i in the sample ($i \in \{1, \dots, p\}$), there are $n_i \in \mathbb{N}$ inter-event times, and

$$\psi_i = ((t_{i,1}, \sigma(I(t_{i,1}))), \dots, (t_{i,n_i}, \sigma(I(t_{i,n_i})))),$$

where $(t_{i,j}, \sigma(I(t_{i,j})))$ contains the j^{th} inter-event time and visited cluster-state ($j \in \{1, \dots, n_i\}$) of the i^{th} element of the sample. Then, the loglikelihood function of the parameters (α_0, D_0, D_1) and sample Ψ is given by the formula

$$\ell(\alpha_0, D_0, D_1; \Psi) = \sum_{i=1}^p \log \left(\alpha_0 \left(\prod_{j=1}^{n_i-1} e^{t_{i,j} D_0} D_1 I_R \right) e^{t_{i,n_i} D_0} D_1 I_{\sigma(I(t_{i,n_i}))} \mathbb{1} \right). \quad (5.1)$$

In the original source, a general version for the CS-MAP is obtained as a direct corollary from the formula for the joint mixed density function of a CS-MAP's first n inter-event times and cluster-states, which we omit, and whose derivation may be found in [4].

Execution time is crucial in the optimization: if the algorithm can look at more parameter sets in less time, it will obtain better estimations in less time. To this effect, there is a trick which decisively accelerates execution time: diagonalizing D_0 . If D_0 is diagonalized as $D_0 = C \Lambda C^{-1}$, then $e^{D_0} = C e^{\Lambda} C^{-1}$ and e^{Λ} is a diagonal matrix, all of which follows from the definition of the matrix exponential. Theoretically, it could be the case that D_0 is not diagonalizable, but in practice this never happens. This means we can always try to apply this trick, and substituting the diagonalized version of the matrix significantly speeds up

execution time. In our case, a speedup of 7 was experienced after introducing the diagonalization trick. Moreover, if parallelization techniques are employed to program the loglikelihood function, an additional speedup of 3 can be obtained.

The first tried approach at RD CS-MAP parameter inference is the maximization of Equation (5.1), and other objective functions derived from it, so this is what is explored in this thesis.

5.2. RD CS-MAP parameter optimization - General procedure

Consider the parameter set of an RD CS-MAP_m, (α_0, D_0, D_1) . There are a total of $2m^2 + m$ components; however, the linear equations $\alpha_0 \mathbb{1} = 1$ and $(D_0 + D_1) \mathbb{1} = 0$ introduce $m + 1$ linear restrictions which reduce the number of actual degrees of freedom of the RD CS-MAP_m to $2m^2 - 1$. Also, considering that the invertibility of D_0 is a given (our D_1 matrices almost never contain crosses of zeroes), we conclude that RD CS-MAP_m optimization, only involves the maximization of a certain objective function of $2m^2 - 1$ variables under a single linear inequality constraint (namely, $\sum_{i=1}^{m-1} \alpha_i \leq 1$) and non-negativity bounds of all variables.

The most obvious idea is to try to maximize the loglikelihood function $\ell(\alpha_0, D_0, D_1)$. We can obtain good approximations to the inter-event time means of the bladder1 dataset by maximizing this function. However, the inter-event time correlations are poorly approximated, so it seems that inter-event time means are approximated at the expense of the inter-event time correlations. Our goal in this thesis, is to achieve a middle ground between both quantities: we will say that a parameter set is a good approximation if it roughly approximates all of the first three conditional inter-event time means and all of the first three conditional inter-event time correlations simultaneously. We used the absolute error metric, because it works the same for numbers in $[0, 1]$ and numbers in $[1, \infty)$.

To this end, we suggest using the following objective function:

$$f(\alpha_0, D_0, D_1 | \Psi) = \ell(\alpha_0, D_0, D_1 | \Psi) - \lambda \sum_{i=1}^2 \sum_{j=i+1}^3 \left(\rho(T_i, T_j | r \geq j-1) - \overline{\rho}_{ij} \right)^2, \quad (5.2)$$

where $\overline{\rho}_{ij}$ denotes the empirical correlations from the dataset, and $\lambda \geq 0$, the penalization parameter, which is a hyperparameter of our choice. This is essentially the loglikelihood function, alongside an L^2 penalization term (like in Ridge Regression) which helps for better approximating the correlations. The reader should note that setting $\lambda = 0$ returns the original loglikelihood function. We will use the function from Equation (5.2) for many values of $\lambda \geq 0$ in the grid

$$\lambda \in \{0, 100, 500, 1000, 5000, 10^4, 5 \cdot 10^4, 10^5, 5 \cdot 10^5, 10^6, 5 \cdot 10^6, 10^7\}$$

for the remainder of this thesis.

For this Master’s Thesis, an optimizer for the estimation of RD CS-MAP parameter sets was designed in Python almost from scratch, using only SciPy’s `optimize.minimize` function. Importantly, we have chosen `trust-constr` as its solver, and the matrix properties of RD CS-MAP parameter sets have been input as `Bounds` and `LinearConstraint` classes, as this was the recommendation from SciPy’s official documentation [25]. The optimization procedure from a given parameter set works as follows:

1. RD CS-MAP parameter sets are fed into a parameters-to-vector function which transforms the matrices into a vector of length $2m^2 - 1$.
2. The new vector and all of the bounds and the linear constraint are fed into the `minimize` function, where the objective function is just the negative of Equation (5.2).
3. After the optimization procedure finishes, the optimal vector is fed into the inverse function of the parameters-to-vector function, the vector-to-parameters function, which returns RD CS-MAP parameters, for easier interpretation.

It has been observed that the loglikelihood function is highly non-convex, so the full algorithm benefits greatly from having a great number of initializations, and a smaller number of gradient descent steps to compensate. Regarding these initializations, we believe that sampling α_0 from a Dirichlet distribution and D_0 off-diagonal elements and D_1 from Gamma distributions would be the simplest most effective way of drawing parameters, at the expense of adding the two Gamma distribution parameters as new possible optimization hyperparameters. We discovered that giving the process many initializations is much more important than applying gradient descent. In the latest versions of our experiment, many parameter sets are sampled but only the top 20 in terms of objective function score are optimized with gradient descent.

Overall, the results that follow were carried out with the following choices of experiment hyperparameters:

- **dataset:** The dataset used for calculating objective function values. It varied from experiment to experiment.
- **order:** The dimensions of the RD CS-MAP. We performed calculations with orders 3, 4, 5.
- **formula:** The objective function. We only used Equation (5.2).
- **initialisations:** The number of gradient descent steps to perform. We fixed it at 10^4 .

- **distribution:** The distribution used for sampling the elements of D_0 and D_1 , which could be uniform or Gamma. In the end, we obtained better results with the Gamma distribution.
- **parameter_A:** The lower bound of the uniform distribution, or the α shape parameter of the Gamma distribution from which matrix elements were sampled. We fixed it at 1.
- **parameter_B:** The upper bound of the uniform distribution, or the θ scale (not rate) parameter of the Gamma distribution from which matrix elements were sampled. We fixed it at 0.5.
- **cost_coefficient:** The penalization term λ in Equation (5.2). We performed calculations with values ranging from 0 to 10^7 .
- **max_SciPy_iterations:** The number of steps in each gradient descent. We fixed it at 20.

Other hyperparameters only influenced numerical precision issues, reproducibility of our results, and control during the executions, which we handled gracefully.

5.3. Optimization on uncensored samples

First, to check that the optimizer works correctly, two different experiments were carried out:

- The estimation of a simulated RD CS-MAP 4000 lifelines sample generated by an order 3 parameter set.
- The estimation of a simulated RD CS-MAP 4000 lifelines sample generated by an order 4 parameter set.

In both cases, estimation was carried out for parameter sets of orders 3, 4 and 5, and parameter sets with reasonably “high” correlations values were chosen for the simulations. Primarily, because it is known that RD CS-MAP estimation can prove to be difficult because correlations are statistically likely to be very close to 0, particularly if the order of the parameters is small and if the death state is absorbing, as is our case (see Chapter 5 from [24] and Chapter 4 from [3]). Therefore, even with the help of the correlations penalization term, we want to make sure that the optimizer searches for RD CS-MAP parameter sets with high correlation values, as inter-event time correlations from lifeline datasets can be high.

Lastly, we applied the optimization procedure on the uncensored individuals of the colorectal dataset only because this dataset contains a relatively small proportion of censored individuals, unlike bladder1.

All of the following values will be truncated at the 4th decimal place, for clarity.

5.3.1. Order 3 Uncensored Simulated RD CS-MAP Sample

For the first sample, the following RD CS-MAP₃ parameter set was used:

$$\alpha_0 = [0.0981 \quad 0.7821 \quad 0.1197], \quad D_0 = \begin{bmatrix} -0.0043 & 0.0001 & 0.0006 \\ 0.0001 & -0.0934 & 0.0000 \\ 0.0018 & 0.0009 & -0.0137 \end{bmatrix}, \quad D_1 = \begin{bmatrix} 0.0028 & 0.0000 & 0.0006 \\ 0.0009 & 0.0564 & 0.0358 \\ 0.0028 & 0.0011 & 0.0069 \end{bmatrix},$$

and it attains the statistics

$$\begin{aligned} \mathbb{E}(T_1 \mid r \geq 0) &= 45.3512 & \rho(T_1, T_2 \mid r \geq 1) &= 0.3649 \\ \mathbb{E}(T_2 \mid r \geq 1) &= 54.8449 & \rho(T_1, T_3 \mid r \geq 2) &= 0.3293 \\ \mathbb{E}(T_3 \mid r \geq 2) &= 62.3542 & \rho(T_2, T_3 \mid r \geq 2) &= 0.3847 \end{aligned}$$

A sample of 4000 instances was generated. With this amount of instances, the effects of the Law of Large Numbers can be clearly seen, and the sample becomes representative of the aforementioned parameters, with values:

$$\begin{aligned} \bar{\mathbb{E}}(T_1 \mid r \geq 0) &= 44.9562 & \bar{\rho}(T_1, T_2 \mid r \geq 1) &= 0.3734 \\ \bar{\mathbb{E}}(T_2 \mid r \geq 1) &= 52.5636 & \bar{\rho}(T_1, T_3 \mid r \geq 2) &= 0.3080 \\ \bar{\mathbb{E}}(T_3 \mid r \geq 2) &= 62.1711 & \bar{\rho}(T_2, T_3 \mid r \geq 2) &= 0.3874 \end{aligned}$$

However, we haven't encountered real life samples with such a large number of individuals.

For the optimization, we used the following hyperparameters:

```
order = 3                                distribution = "gamma"
formula = "pepa"                         parameter_A = 1
initialisations = 10**4                  parameter_B = 0.5
max_SciPy_iterations = 5                 cost_coefficient = [varying values]
```

The results from the optimization of the original loglikelihood can be consulted in Table 5.1. Testing the different orders, it can be seen that the best parameter sets from all orders perform well, all surpassing the original parameter set's loglikelihood. In the results from the optimization, the top 20 original parameters may be consulted, and it can be seen that most of them obtain similar loglikelihoods to the original parameter set.

However, all approximations to either the empirical and original inter-event time correlations turned out to be off by roughly 20 units for the inter-event time means and 0.15 for the inter-event time correlations.

Table 5.1. Estimates of the order 3 dataset - Comparison of inter-event time means and inter-event time correlations between different orders.

\	Original	Empirical	Order 3 Estimation	Order 4 Estimation	Order 5 Estimation
Log-likelihood	-53215	-	-49100	-49277	-49281
$\mathbb{E}(T_1 r \geq 0)$	45.35	44.95	28.78	28.33	34.95
$\mathbb{E}(T_2 r \geq 1)$	54.84	52.56	31.85	33.26	45.27
$\mathbb{E}(T_3 r \geq 2)$	62.35	62.17	33.13	35.15	52.24
$\rho(T_1, T_2 r \geq 1)$	0.3649	0.3734	0.2308	0.2557	0.2136
$\rho(T_1, T_3 r \geq 2)$	0.3293	0.3080	0.1318	0.1637	0.1091
$\rho(T_2, T_3 r \geq 2)$	0.3847	0.3874	0.2409	0.2663	0.2111

In Figure 5.1, after fixing for order 3 parameter sets, constrained optimization was enforced, for a wide range of penalization term values. The end result is that for most cost penalization terms, the error in the approximation of the means varies unpredictably, while the error in the approximation of the correlations decreases steadily. Therefore, for this dataset, the best parameters are the optimal parameters from the optimization with $\lambda = 5 \cdot 10^4$ (or other nearby values), which were

$$\widehat{\alpha}_0 = [0.0991 \quad 0.8129 \quad 0.0878], \quad \widehat{D}_0 = \begin{bmatrix} -0.0052 & 0.0000 & 0.0000 \\ 0.0055 & -0.1985 & 0.1011 \\ 0.0074 & 0.0436 & -0.1647 \end{bmatrix}, \quad \widehat{D}_1 = \begin{bmatrix} 0.0027 & 0.0000 & 0.0023 \\ 0.0001 & 0.0403 & 0.0513 \\ 0.0008 & 0.0863 & 0.0264 \end{bmatrix},$$

attaining the statistics

$$\begin{aligned} \widehat{\mathbb{E}}(T_1 | r \geq 0) &= 37.64 & \widehat{\rho}(T_1, T_2 | r \geq 1) &= 0.3429 \\ \widehat{\mathbb{E}}(T_2 | r \geq 1) &= 45.54 & \widehat{\rho}(T_1, T_3 | r \geq 2) &= 0.2852 \\ \widehat{\mathbb{E}}(T_3 | r \geq 2) &= 52.83 & \widehat{\rho}(T_2, T_3 | r \geq 2) &= 0.3469 \end{aligned}$$

Execution time for all experiments regarding this sample took roughly 33 hours. We believe this is simply because a sample of 4000 patients is big for our program.

5.3.2. Order 4 Uncensored Simulated RD CS-MAP Sample

For the second sample, the following RD CS-MAP₄ parameter set was used:

$$\alpha_0 = [0.2174 \quad 0.1963 \quad 0.4403 \quad 0.1458], \quad D_0 = \begin{bmatrix} -2.0229 & 0.0011 & 0.1746 & 0.0106 \\ 0.0322 & -0.2144 & 0.0012 & 0.0216 \\ 0.3683 & 0.4703 & -2.9485 & 0.1548 \\ 0.0108 & 1.2305 & 0.5657 & -3.1013 \end{bmatrix}, \quad D_1 = \begin{bmatrix} 1.2500 & 0.0134 & 0.3889 & 0.0184 \\ 0.0002 & 0.1190 & 0.0386 & 0.0014 \\ 1.3792 & 0.0542 & 0.1096 & 0.4119 \\ 0.0058 & 0.2822 & 0.1553 & 0.8507 \end{bmatrix},$$

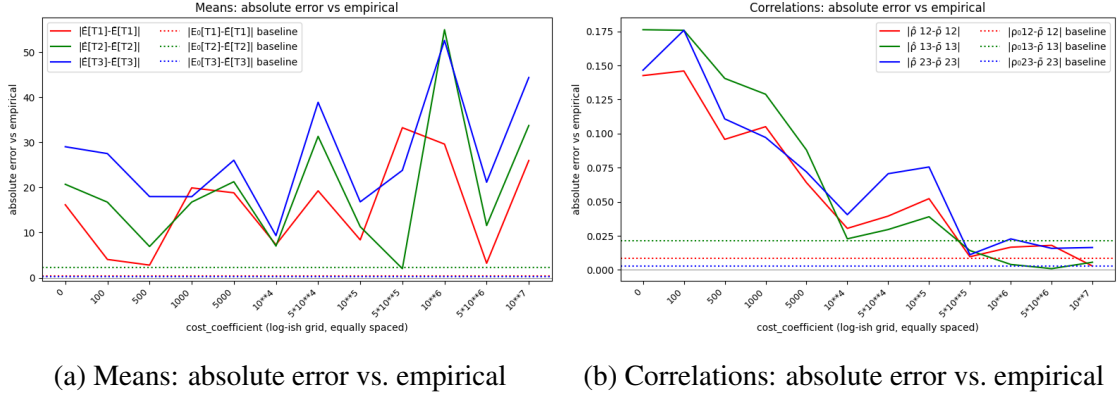


Fig. 5.1. Uncensored order 3 sample - model performance across penalization terms.

attaining the statistics

$$\begin{aligned} \mathbb{E}(T_1 | r \geq 0) &= 2.0932 & \rho(T_1, T_2 | r \geq 1) &= 0.2052 \\ \mathbb{E}(T_2 | r \geq 1) &= 1.9804 & \rho(T_1, T_3 | r \geq 2) &= 0.1382 \\ \mathbb{E}(T_3 | r \geq 2) &= 1.7719 & \rho(T_2, T_3 | r \geq 2) &= 0.2252 \end{aligned}$$

As can be seen, inter-event time means for this parameter set are significantly smaller. Moreover, it may be the case that CS-MAP parameter sets with low inter-event time means are more common than those with high inter-event time means, just like with the correlations, so we expected representative samples from this parameter set to be better approximated. The inter-event time correlations were also chosen to be smaller, but not negligible.

A sample of 4000 instances was generated, with statistics:

$$\begin{aligned} \bar{\mathbb{E}}(T_1 | r \geq 0) &= 2.0339 & \bar{\rho}(T_1, T_2 | r \geq 1) &= 0.1946 \\ \bar{\mathbb{E}}(T_2 | r \geq 1) &= 1.9559 & \bar{\rho}(T_1, T_3 | r \geq 2) &= 0.1213 \\ \bar{\mathbb{E}}(T_3 | r \geq 2) &= 1.6262 & \bar{\rho}(T_2, T_3 | r \geq 2) &= 0.2347 \end{aligned}$$

For the optimization, the following experiment hyperparameters were used:

order = 4	distribution = "gamma"
formula = "pepa"	parameter_A = 1
initialisations = 10**4	parameter_B = 0.5
max_SciPy_iterations = 5	cost_coefficient = [varying values]

The results from the optimization of the original loglikelihood can be consulted in Table 5.2. Testing the different orders, we see that not only all perform well, but also that the order 3 parameter set, the one with the least free parameters, is the best performing in terms of approximating the inter-event time means and correlations. Thus, since the three

parameter sets have very similar loglikelihood scores, this suggests that loglikelihood is not the best scoring method for optimizing inter-event time means and correlations approximations.

Table 5.2. Estimates of the order 4 dataset - Comparison of inter-event time means and inter-event time correlations between different orders.

\	Original	Empirical	Order 3 Estimation	Order 4 Estimation	Order 5 Estimation
Log-likelihood	-47965	-	-46019	-46186	-46003
$\mathbb{E}(T_1 r \geq 0)$	2.0932	2.0339	2.4898	1.9627	1.7337
$\mathbb{E}(T_2 r \geq 1)$	1.9804	1.9559	2.1111	2.0544	1.6417
$\mathbb{E}(T_3 r \geq 2)$	1.7719	1.6262	1.9035	1.5872	1.4369
$\rho(T_1, T_2 r \geq 1)$	0.2052	0.1946	0.2239	0.1551	0.1863
$\rho(T_1, T_3 r \geq 2)$	0.1382	0.1213	0.1540	0.0634	0.1020
$\rho(T_2, T_3 r \geq 2)$	0.2252	0.2347	0.2397	0.1781	0.2275

In Figure 5.2, after fixing for order 4 parameter sets, constrained optimization was enforced. The end result is that unconstrained optimization already yields a decent approximation of the correlations (with errors of ± 0.05), but also the most precise mean approximations. However, we believed the best parameter set to be the one associated to $\lambda = 500$, since it approximates the means and correlations well simultaneously. Its associated parameter set is

$$\widehat{\alpha}_0 = [0.2526 \ 0.3710 \ 0.3035 \ 0.0728], \quad \widehat{D}_0 = \begin{bmatrix} -0.2137 & 0.0040 & 0.0006 & 0.0018 \\ 0.1695 & -2.0470 & 0.1655 & 0.0174 \\ 0.0691 & 0.0255 & -2.0625 & 0.0215 \\ 0.0041 & 0.1428 & 0.0822 & -0.7760 \end{bmatrix}, \quad \widehat{D}_1 = \begin{bmatrix} 0.1207 & 0.0086 & 0.0590 & 0.0188 \\ 0.0236 & 1.1188 & 0.4432 & 0.1087 \\ 0.0157 & 0.0277 & 1.6117 & 0.2910 \\ 0.1563 & 0.2573 & 0.0180 & 0.1149 \end{bmatrix},$$

attaining the statistics

$$\begin{aligned} \widehat{\mathbb{E}}(T_1 | r \geq 0) &= 1.8582 & \widehat{\rho}(T_1, T_2 | r \geq 1) &= 0.2044 \\ \widehat{\mathbb{E}}(T_2 | r \geq 1) &= 1.6200 & \widehat{\rho}(T_1, T_3 | r \geq 2) &= 0.1287 \\ \widehat{\mathbb{E}}(T_3 | r \geq 2) &= 1.4364 & \widehat{\rho}(T_2, T_3 | r \geq 2) &= 0.2274 \end{aligned}$$

Total execution time of the experiments regarding this sample took roughly 40 hours. It is similar to the order 3's sample, and we believe the extra time is only attributed to the higher order of the parameter sets.

5.3.3. colorectal

Lastly, we applied our developed inference techniques to analyze the colorectal dataset. We recall the statistics we are trying to estimate for the dataset without the censored indi-

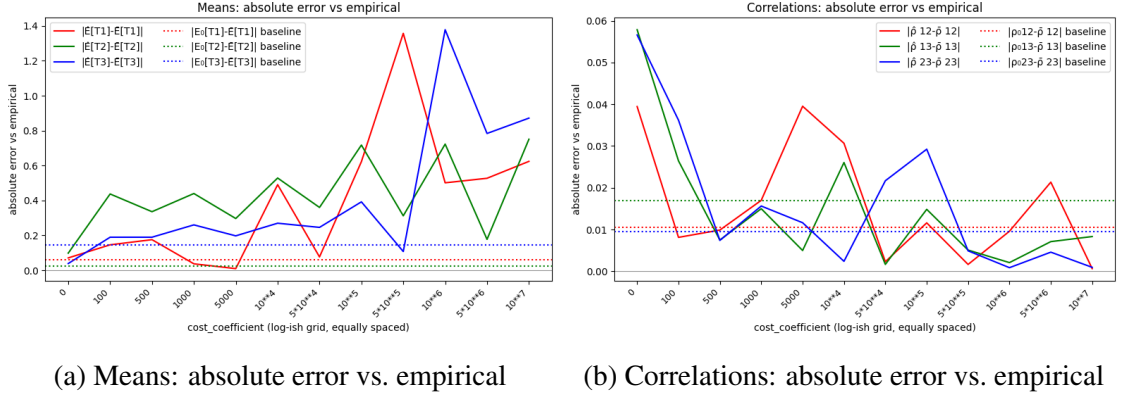


Fig. 5.2. Uncensored order 4 sample - model performance across penalization terms.

viduals:

$$\begin{aligned}
\bar{\mathbb{E}}(T_1 \mid r \geq 0) &= 0.6849 & \bar{\rho}(T_1, T_2 \mid r \geq 1) &= 0.3543 \\
\bar{\mathbb{E}}(T_2 \mid r \geq 1) &= 0.5056 & \bar{\rho}(T_1, T_3 \mid r \geq 2) &= 0.0799 \\
\bar{\mathbb{E}}(T_3 \mid r \geq 2) &= 0.3796 & \bar{\rho}(T_2, T_3 \mid r \geq 2) &= 0.0934.
\end{aligned}$$

As it turned out, by removing from the dataset the 20% of censored individuals, we have an optimization with a reasonably accurate result, that being, the one with $(m, \lambda) = (4, 10^4)$. The parameter set is

$$\hat{\alpha}_0 = \begin{bmatrix} 0.0204 & 0.9761 & 0.0017 & 0.0015 \end{bmatrix}, \quad \hat{D}_0 = \begin{bmatrix} -1.2226 & 0.0669 & 0.1025 & 0.7509 \\ 0.6191 & -2.8762 & 0.4247 & 0.2796 \\ 0.0006 & 0.0001 & -4.0979 & 0.0020 \\ 0.0080 & 0.0047 & 0.0956 & -1.2477 \end{bmatrix}, \quad \hat{D}_1 = \begin{bmatrix} 0.0809 & 0.0021 & 0.0001 & 0.2190 \\ 0.0010 & 0.0098 & 1.2878 & 0.2539 \\ 0.8492 & 0.1937 & 0.4384 & 2.6137 \\ 0.7504 & 0.0014 & 0.0017 & 0.3857 \end{bmatrix},$$

which attained the values

$$\begin{aligned}
\widehat{\mathbb{E}}(T_1 \mid r \geq 0) &= 0.7762 & \widehat{\rho}(T_1, T_2 \mid r \geq 1) &= 0.3455 \\
\widehat{\mathbb{E}}(T_2 \mid r \geq 1) &= 0.5938 & \widehat{\rho}(T_1, T_3 \mid r \geq 2) &= 0.0773 \\
\widehat{\mathbb{E}}(T_3 \mid r \geq 2) &= 1.0771 & \widehat{\rho}(T_2, T_3 \mid r \geq 2) &= 0.1036,
\end{aligned}$$

and a score of -383.8 .

From the results, we can see that the first three inter-event times means are well approximated up to an error of ± 0.1 , except for the third one; and simultaneously, the correlations are successfully approximated almost up to an error of ± 0.01 . All of the results may be consulted in Table 5.3., and as can be seen, other choices of (m, λ) yielded much more inaccurate approximations.

Table 5.3. Best estimation results for the colorectal dataset.

Metric	Empirical	$m = 3$			$m = 4$			$m = 5$		
		$\lambda = 0$	$\lambda = 100$	$\lambda = 10^4$	$\lambda = 0$	$\lambda = 100$	$\lambda = 10^4$	$\lambda = 0$	$\lambda = 100$	$\lambda = 10^4$
Score	—	-359.3	-360.5	-370.5	-359.8	-372.8	-383.8	-365.3	-380.3	-570.0
$\mathbb{E}(T_1 r \geq 0)$	0.6849	1.0129	1.0901	1.1440	1.0739	1.0575	0.7762	1.0120	0.9963	0.8576
$\mathbb{E}(T_2 r \geq 1)$	0.5056	0.6000	0.6567	0.6146	0.5966	0.6010	0.5938	0.6421	0.6321	0.6072
$\mathbb{E}(T_3 r \geq 2)$	0.3796	0.5652	0.5512	1.2601	0.5340	0.5374	1.0771	0.6440	0.6383	0.6736
$\rho(T_1, T_2 r \geq 1)$	0.3543	0.0015	0.1948	0.3503	-0.0031	0.0100	0.3455	0.0003	0.0019	0.2515
$\rho(T_1, T_3 r \geq 2)$	0.0799	0.0002	0.0720	0.0861	0.0004	-0.0014	0.0773	0.0000	0.0000	0.1274
$\rho(T_2, T_3 r \geq 2)$	0.0934	0.0017	0.1936	0.0841	0.0185	0.0241	0.1036	-0.0008	-0.0019	0.1456

5.4. Optimization on censored samples

After testing the optimization program on uncensored, simulated datasets, our next goal was to analyze the effects of censorship on estimation. Before delving into the results, the processes of handling the censored data must be explained.

We dealt with censorship in 3 different ways:

- By dropping all censored samples and reducing our dataset.
- By working with a loglikelihood function and empirical statistics functions adapted for censored samples (see Chapter 5.2.1. in [4]).
- By imputing data by the methodology described in Section 2.3.3. first for the r variable and then for the times variables.

Since we expected the first method to perform poorly, we only implemented it against a simulated sample. The other two methods were put to the test with both the simulated sample and the bladder1 cancer dataset.

5.4.1. Order 3 Censored Simulated RD CS-MAP Sample

A censored sample of 200 individuals was generated from the same parameter set as the one from the order 3 uncensored simulated sample.

The way the censored sample was generated is as follows: first, an uncensored sample was generated, using the previously defined Python functions. Afterwards, we implemented a censoring mechanism that performed the following:

1. Randomly choose as many individuals to censor as desired.

2. For each chosen individual, censor its r variable with a uniform distribution of lower bound 0 and upper bound the original value.
3. For each censored individual, remove any times with index higher than the new r value resulting from censorship, and censor the last uncensored time with an exponential distribution of rate defined by the user.

For our experiment, we censored 75% of the individuals from the sample, and chose an exponential distribution of rate 0.1. In general, the method of generating censored lifelines datasets is very challenging because it involves choosing and calibrating probability distributions in a way that is adequate to a particular dataset. Moreover, in the case of lifelines datasets with multiple events, a choice must be made of whether to censor the r variable or not. For a rigorous approach of the censorship mechanism of lifelines datasets with single events [26] is recommended.

For the first method, working with the only 50 uncensored individuals, the empirical statistics of the uncensored group are:

$$\begin{aligned}\bar{E}(T_1 | r \geq 0) &= 45.00 & \bar{\rho}(T_1, T_2 | r \geq 1) &= 0.2532 \\ \bar{E}(T_2 | r \geq 1) &= 47.96 & \bar{\rho}(T_1, T_3 | r \geq 2) &= 0.2616 \\ \bar{E}(T_3 | r \geq 2) &= 72.04 & \bar{\rho}(T_2, T_3 | r \geq 2) &= 0.3026,\end{aligned}$$

which are somewhat different to the original parameter set's results, but still representative of the original parameter set.

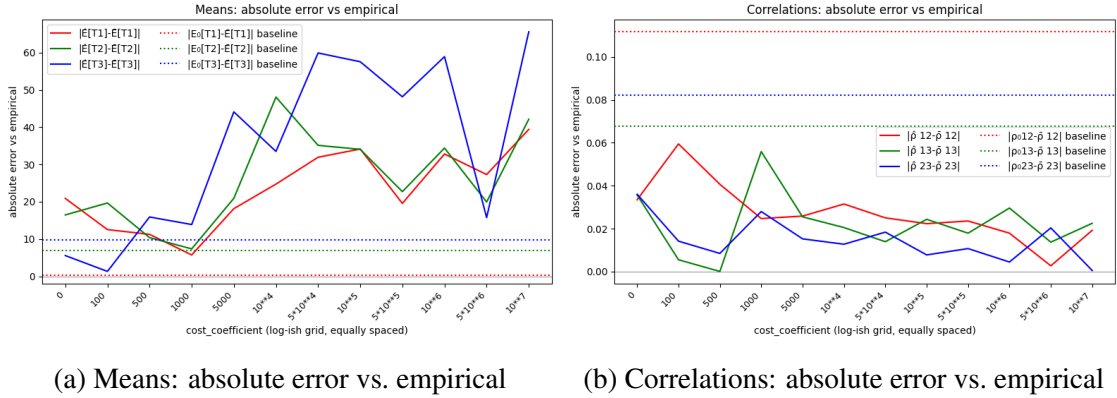


Fig. 5.3. Censored order 3 sample - model performance across penalization terms, ignoring censored individuals.

The best parameter set was chosen to be the one associated to $\lambda = 1000$, which obtained the statistics

$$\begin{aligned}\widehat{E}(T_1 | r \geq 0) &= 50.74 & \widehat{\rho}(T_1, T_2 | r \geq 1) &= 0.2779 \\ \widehat{E}(T_2 | r \geq 1) &= 55.33 & \widehat{\rho}(T_1, T_3 | r \geq 2) &= 0.2057 \\ \widehat{E}(T_3 | r \geq 2) &= 58.14 & \widehat{\rho}(T_2, T_3 | r \geq 2) &= 0.2746.\end{aligned}$$

For the second method, dealing with censored values with a modified loglikelihood function and empirical statistics functions, the results we obtained were fine. For our best approximation, the one associated to $\lambda = 100$, we obtained the statistics

$$\begin{aligned}\widehat{\mathbb{E}}(T_1 | r \geq 0) &= 58.68 & \widehat{\rho}(T_1, T_2 | r \geq 1) &= 0.3013 \\ \widehat{\mathbb{E}}(T_2 | r \geq 1) &= 66.31 & \widehat{\rho}(T_1, T_3 | r \geq 2) &= 0.2529 \\ \widehat{\mathbb{E}}(T_3 | r \geq 2) &= 68.85 & \widehat{\rho}(T_2, T_3 | r \geq 2) &= 0.3025.\end{aligned}$$

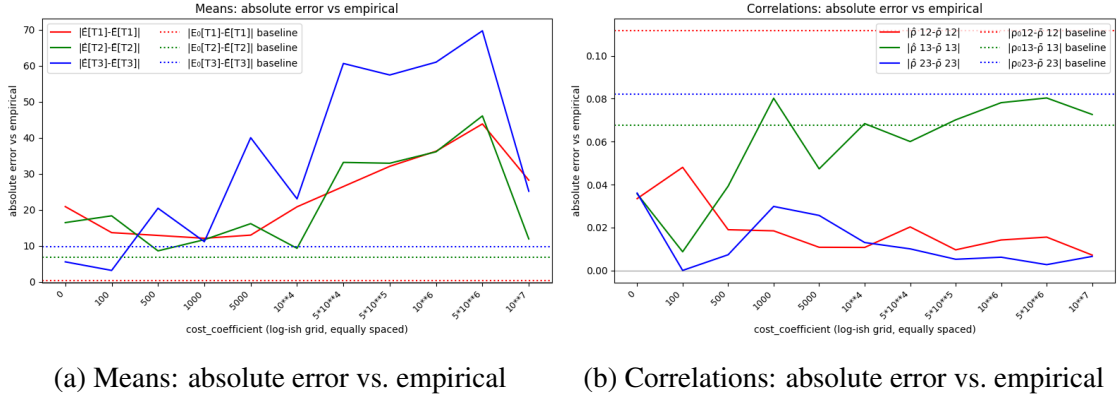


Fig. 5.4. Censored order 3 sample - model performance across penalization terms, using a modified loglikelihood function.

Lastly, for the third method, by replacing censored data with newly imputed data, we obtained some decent approximations. Our new empirical statistics became

$$\begin{aligned}\bar{\mathbb{E}}(T_1 | r \geq 0) &= 63.24 & \bar{\rho}(T_1, T_2 | r \geq 1) &= 0.2301 \\ \bar{\mathbb{E}}(T_2 | r \geq 1) &= 53.55 & \bar{\rho}(T_1, T_3 | r \geq 2) &= 0.1943 \\ \bar{\mathbb{E}}(T_3 | r \geq 2) &= 76.60 & \bar{\rho}(T_2, T_3 | r \geq 2) &= 0.2764,\end{aligned}$$

and for $\lambda = 5000$, our best approximation, we obtained

$$\begin{aligned}\widehat{\mathbb{E}}(T_1 | r \geq 0) &= 54.92 & \widehat{\rho}(T_1, T_2 | r \geq 1) &= 0.1807 \\ \widehat{\mathbb{E}}(T_2 | r \geq 1) &= 62.57 & \widehat{\rho}(T_1, T_3 | r \geq 2) &= 0.1489 \\ \widehat{\mathbb{E}}(T_3 | r \geq 2) &= 64.56 & \widehat{\rho}(T_2, T_3 | r \geq 2) &= 0.1805.\end{aligned}$$

5.4.2. bladder1

Lastly, regarding the bladder1 dataset, our first approach involved the use of the loglikelihood function, which yielded unsatisfactory results, and can be visualized in Figure 5.3. Note, we skipped plotting the results for $(m, \lambda) \in \{(3, 10^6), (4, 100)\}$ because they were very inaccurate and severely distorted the line graphs for the rest of the pairs.

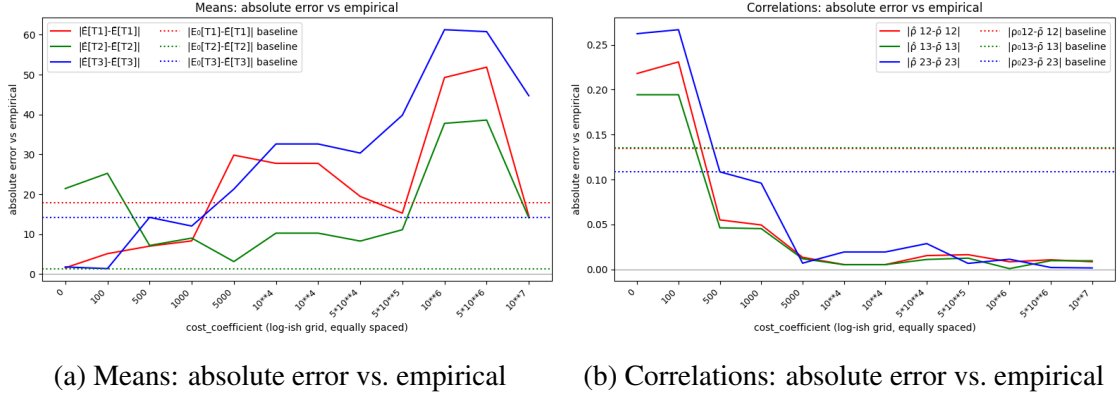


Fig. 5.5. Censored order 3 sample - model performance across penalization terms, imputing data for censored individuals.

The best parameter set was the one associated to $(m, \lambda) = (3, 10^5)$, which was

$$\widehat{\alpha}_0 = \begin{bmatrix} 0.0000 & 0.4203 & 0.5795 \end{bmatrix}, \quad \widehat{D}_0 = \begin{bmatrix} -0.0620 & 0.0004 & 0.0507 \\ 0.0000 & -0.4038 & 0.0008 \\ 0.0000 & 0.1529 & -1.3600 \end{bmatrix}, \quad \widehat{D}_1 = \begin{bmatrix} 0.0070 & 0.0000 & 0.0037 \\ 0.0001 & 0.3810 & 0.0218 \\ 0.6115 & 0.0051 & 0.5902 \end{bmatrix},$$

attaining the statistics

$$\begin{aligned} \widehat{E}(T_1 | r \geq 0) &= 1.6302 & \widehat{\rho}(T_1, T_2 | r \geq 1) &= -0.2254 \\ \widehat{E}(T_2 | r \geq 1) &= 7.7243 & \widehat{\rho}(T_1, T_3 | r \geq 2) &= -0.1830 \\ \widehat{E}(T_3 | r \geq 2) &= 5.6013 & \widehat{\rho}(T_2, T_3 | r \geq 2) &= 0.3092, \end{aligned}$$

and a score of -2175 .

Clearly, it fails to approximate the first inter-event time mean, and also doesn't fully capture the inter-event time correlations, even though it approximates the second and third inter-event time means much better.

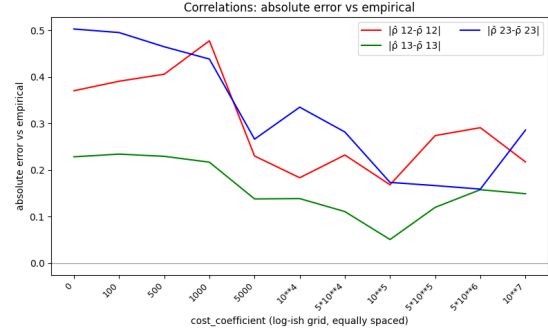
For the second method, involving data imputation, we also obtained unsatisfactory results, even after the empirical statistics changed to

$$\begin{aligned} \bar{E}(T_1 | r \geq 0) &= 12.2901 & \bar{\rho}(T_1, T_2 | r \geq 1) &= -0.2518 \\ \bar{E}(T_2 | r \geq 1) &= 8.1022 & \bar{\rho}(T_1, T_3 | r \geq 2) &= -0.1040 \\ \bar{E}(T_3 | r \geq 2) &= 7.3860 & \bar{\rho}(T_2, T_3 | r \geq 2) &= 0.3715. \end{aligned}$$

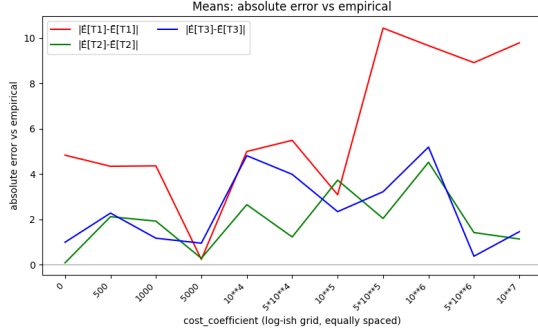
The graphs can be consulted in Figure 5.4.



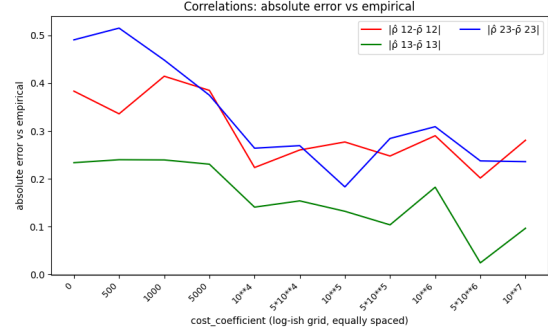
(a) Order 3 — means



(b) Order 3 — correlations

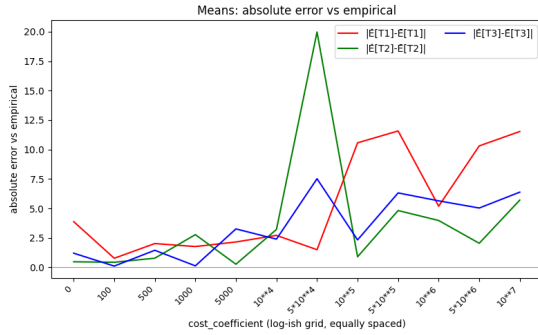


(c) Order 4 — means

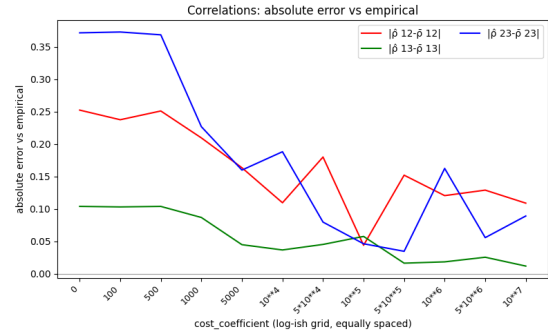


(d) Order 4 — correlations

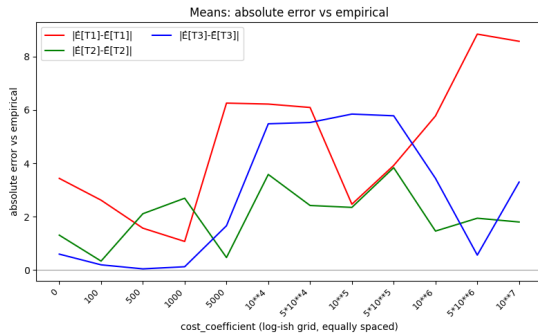
Fig. 5.6. Estimated means (left) and correlations (right) for CS-MAP orders 3 and 4 estimations of bladder1, with a censorship objective function.



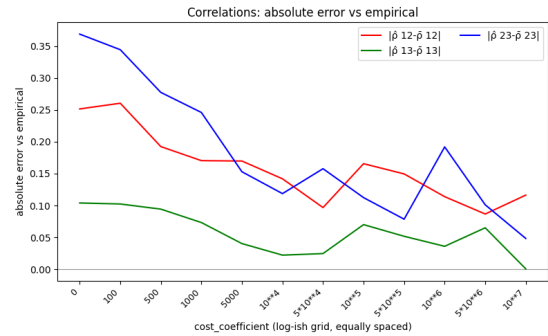
(a) Order 3 — means



(b) Order 3 — correlations



(c) Order 4 — means



(d) Order 4 — correlations

Fig. 5.7. Estimated means (left) and correlations (right) for CS-MAP orders 3 and 4 estimations of bladder1, with censorship imputed.

The best suited dataset can be argued to be the one associated to $(m, \lambda) = (4, 5 \cdot 10^6)$, with parameter set

$$\widehat{\alpha}_0 = \begin{bmatrix} 0.0229 & 0.1572 & 0.0101 & 0.8097 \end{bmatrix}, \quad \widehat{D}_0 = \begin{bmatrix} -0.0670 & 0.0000 & 0.0005 & 0.0021 \\ 0.0008 & -0.1838 & 0.0685 & 0.0027 \\ 0.0021 & 0.0381 & -0.7774 & 0.0156 \\ 0.0028 & 0.0975 & 0.0649 & -0.6710 \end{bmatrix}, \quad \widehat{D}_1 = \begin{bmatrix} 0.0236 & 0.0003 & 0.0003 & 0.0400 \\ 0.0015 & 0.0468 & 0.0474 & 0.0158 \\ 0.0000 & 0.0136 & 0.6942 & 0.0136 \\ 0.4235 & 0.0191 & 0.0154 & 0.0476 \end{bmatrix},$$

and a score of -111779 .

6. CONCLUSIONS

In this Master’s thesis, results about the theory of RD CS-MAPs and the bladder1 and colorectal lifelines datasets have been explored.

Regarding MAPs, we supported with a mathematical argument, in the form of a proven theorem, why RD CS-MAP optimizers (and MAP or general CS-MAP optimizers) generally won’t need to check the invertibility condition of D_0 . Its proof uses the Perron-Frobenius Theorem at its core, and can be consulted in the Appendix.

Regarding RD CS-MAPs, we derived a formula for the moments of the number of recurrences variable, $\mathbb{E}(r^k)$ for any $k \in \mathbb{N}$. The formula was proven rigorously using two key results: the formula for the generating function of the sequence $\{n^k\}_{n=1}^{\infty}$, and power series theory from Functional Analysis for the convergence aspects. On top of these, several standard combinatorial results were used and mathematical manipulations carried out. The details of the proof may be consulted in the Appendix. Moreover, we also justified with another mathematical argument why $\mathbb{E}(r)$ is representative of r ’s distribution: r ’s distribution is uniquely defined by its sequence of moments.

Moving onto RD CS-MAP experiments, first we investigated whether the absolute values of inter-event time correlations had an effect on the distribution of $\mathbb{E}(r)$ and $\sqrt{\mathbb{V}(r)}$, which are representative of r , after fixing the first three inter-event time means, and could have only been feasibly carried out with the previously derived formula. The distribution of (Percentile 95)(r) was also analyzed, for seeing the impact of the correlations on the *tail* of r ’s distribution. However, these experiments were not conclusive, and we would have had to enforce much stricter criteria for “high” correlations parameter sets at the expense of much more computational time in order to see new effects. Changing the sampling distribution is another possible change which could also be carefully investigated.

A second experiment about RD CS-MAP was performed. The aim of the experiment was to discern whether the absolute values of inter-event time correlations had an effect on the distribution of the death time T_D , conditioned on the fact that T_1 was larger than some threshold t . As representatives of $T_D|_{T_1 > t}$, we chose its mean, median and 95th percentile. Theoretical formulas for the mean, median and 95th percentile were not available to us, (not even implicit formulas), so MonteCarlo simulations for estimating these had to be carefully planned out and performed, at the expense of both development and execution time. However, despite these issues, we were able to successfully conclude that high (in absolute value) correlations do have an effect on the conditional death time’s distribution. The conditional death time may occasionally reach very large values if the inter-event

time correlations are high. If this experiment was to be carried out again, it would be better if formulas for the conditional moments and conditional distribution functions of T_D could be derived, and if stricter “high” correlations thresholds could be set. Changing the sampling distribution should also be carefully investigated.

In the last section, we briefly reviewed the known results about CS-MAP parameter inference, and we designed an optimizer in Python to this effect. Modeling datasets by focusing on correctly estimating the inter-event time correlations is a new approach that has not been explored extensively. We discovered that, generally, the loglikelihood function suffices for approximating the inter-event time means, but an extra modification was needed in order to correctly approximate the inter-event time correlations if the dataset featured high correlations. To approximate the inter-event time correlations, the objective function was modified: a quadratic penalization term regarding the inter-event time correlations was subtracted from the loglikelihood function. This way, coupling a large number of initializations alongside some gradient descent steps to our best parameter sets, we obtained results which approximated the inter-event time means and correlations with varying degrees of success.

First, uncensored samples were explored, including samples from artificial simulations and the uncensored samples from the real life datasets `bladder1` and `colorectal`. In the case of the simulated samples, we concluded that penalizing the inter-event time correlations was a successful approach at obtaining RD CS-MAP parameter sets which correctly estimated the correlations without worsening the inter-event time mean estimates too much. In the case of `bladder1`, we concluded that dropping the censored patients was not good, since this left us with a dataset too small to be used for Statistics. However, since the `colorectal` dataset featured comparatively fewer censored patients, we were successful at approximating the inter-event time means up to a degree of accuracy of ± 0.1 and the inter-event time correlations up to a degree of accuracy of ± 0.01 , with a single RD CS-MAP parameter set.

Lastly, we explored censored samples, as well as the censoring procedures themselves. There is no standard way of correctly censoring lifelines datasets with more than one type of event, so we used three different methods for estimating the simulated samples: dropping censored samples, using a modified loglikelihood function coupled with Kaplan-Meier empirical statistics functions for censored samples, and a method of data imputation based on Kaplan-Meier estimation. In the end, all methods obtained performed similarly with the simulated sample. Lastly, the last two approaches were used for modeling the `bladder1` dataset, however, the obtained approximations did not model the dataset accurately. We believe that employing a new objective function with a quadratic penalization term based on the inter-event time means should increase the accuracy of the inter-event time means.

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APPENDIX

In the Appendix, we provide the proofs of some of the theorems claimed in this thesis.

First, we provide a proof of Theorem 1, regarding the relationship between the invertibility of D_0 and the elements of D_1 , where D_0 and D_1 are MAP parameters.

Proof.

First, suppose that D_1 has no rows of all zeroes. From the first remark, we know that $(D_0 + D_1)\mathbb{1} = 0$, so $D_0\mathbb{1} = -D_1\mathbb{1}$. Since D_1 is a non-negative matrix with no rows of zeroes and $\mathbb{1}$ is a positive vector, $D_1\mathbb{1}$ must be a positive vector, thus $D_0\mathbb{1}$ is a negative vector. From this, we may use the following fact, derived from the Perron-Frobenius Theorem and proved in a research paper [33, Corollary 8.2] (we cite it down below just for convenience):

“Let A be a matrix with negative diagonal elements and non-negative off-diagonal elements, v be a positive vector, and λ be a non-negative real number such that $Av - \lambda v$ is a negative vector. Then $\eta(A) < \lambda$, where $\eta(A) = \max\{\text{Re}(\mu) \mid \mu \text{ eigenvalue of } A\}$.”

Taking $\lambda \equiv 0$, $A \equiv D_0$, and $v \equiv \mathbb{1}$ in the previous theorem, the hypotheses hold, so we obtain that $\eta(D_0) < 0$, meaning that all eigenvalues of D_0 have a negative real part. In particular, 0 is not an eigenvalue of D_0 , so D_0 is invertible, as we wanted to show.

Lastly, suppose that D_1 has no columns of all zeroes. Then, D_1^T has no rows of all zeroes; and D_0^T and D_1^T satisfy the same properties as in the previous case. Following the exact same logic with D_0^T and D_1^T , we deduce that 0 is not an eigenvalue of D_0^T , and since every square matrix shares the same eigenvalues as their transpose, 0 is not an eigenvalue of D_0 and so D_0 is invertible. $\square$

Next is a proof of Theorem 3, the theoretical formula for the moments of the r variable of an RD CS-MAP, the number of recurrences until death, as well as the fact that its distribution solves the “moments problem”.

Proof.

For this proof, we need to state some auxiliary results. First, we recall some elementary combinatorial formulas about Stirling numbers of the second kind:

- $\forall k \in \mathbb{N}, j \in \{1, \dots, k\}$:

$$\sum_{i=j}^k \binom{k}{i} \left\{ \begin{matrix} i \\ j \end{matrix} \right\} = \left\{ \begin{matrix} k+1 \\ j+1 \end{matrix} \right\}, \quad (6.1)$$

from which it is immediately deduced that

$$\sum_{i=j}^{k-1} \binom{k}{i} \binom{i}{j} = \binom{k+1}{j+1} - \binom{k}{j}, \quad (6.2)$$

by recalling that $\binom{k}{k} = 1$ and subtracting the k^{th} summand from both sides of (4.2).

- $\forall k \in \mathbb{N}, j \in \{1, \dots, k\}$:

$$\binom{k+1}{j} = j \binom{k}{j} + \binom{k}{j-1}, \quad (6.3)$$

from which it is immediately deduced that

$$\binom{k+1}{j+1} - \binom{k}{j} = (j+1) \binom{k}{j+1}, \quad (6.4)$$

by performing the change of variables $j \rightarrow j+1$ and subtracting the second summand from both sides of (4.4).

Next, we state the formula for the generating function of the sequence n^k , for all $k \in \mathbb{N}$:

$$\sum_{n=0}^{\infty} n^k x^n = \frac{1}{1-x} \sum_{i=0}^k i! \binom{k}{i} \frac{x^i}{(1-x)^i}, \quad (6.5)$$

with convergence only for values of $x \in \mathbb{R}$ such that $|x| < 1$ [30, equations (3.3) and (3.7)].

This takes care of the symbolic formula, but not of the convergence aspects within the Banach algebra of square matrices. Even though the original paper only references real-number arguments, it can be quickly generalized into a result about matrices, using the fact that the series is just a power series over the unital Banach algebra of complex matrices [32]. The only matrices for which it converges are matrices A such that $\rho(A) < 1$, where $\rho(\cdot)$ is the *spectral radius* of a matrix (i.e. the maximum modulus of its eigenvalues). This is akin to the geometric series formula for matrices, which imposes the same hypothesis on matrices (and which was already used to derive the formula for $\mathbb{E}(r)$ in [4]).

All in all, the result can be summarised as follows: for all $k \in \mathbb{N}$:

$$\sum_{n=0}^{\infty} n^k A^n = (I - A)^{-1} \sum_{i=0}^k i! \binom{k}{i} A^i (I - A)^{-i}, \quad (6.6)$$

with convergence only for values of $A \in \mathbb{R}^{m \times m}$ such that $\rho(A) < 1$.⁴

⁴Since A and $(I - A)^{-1}$ commute, we will write $A^i (I - A)^{-i} = (A(I - A)^{-1})^i$.

Next, we check the formula for $\mathbb{E}(r^k)$. Denote $A \equiv P^* I_R$. Already knowing that $\rho(A) < 1$:

$$\begin{aligned}\mathbb{E}(r^k) &= \sum_{n=0}^{\infty} \mathbb{P}(r^k > n) \\ &= \sum_{n=0}^{\infty} \mathbb{P}(r > \sqrt[k]{n}) \\ &= \sum_{n=0}^{\infty} \mathbb{P}(r > \lfloor \sqrt[k]{n} \rfloor) \\ &= \sum_{n=0}^{\infty} \alpha_0 A^{\lfloor \sqrt[k]{n} \rfloor + 1} \mathbb{1}, \quad \text{by Theorem 1}\end{aligned}$$

$$\begin{aligned}&= \alpha_0 A \left(\sum_{n=0}^{\infty} A^{\lfloor \sqrt[k]{n} \rfloor} \right) \mathbb{1} \\ &= \alpha_0 A \left(\sum_{a=0}^{\infty} ((a+1)^k - a^k) A^a \right) \mathbb{1},\end{aligned}$$

by noting that for all $a \in \mathbb{N}_{\geq 0}$: $a = \lfloor \sqrt[k]{n} \rfloor \iff a \leq \lfloor \sqrt[k]{n} \rfloor < a+1 \iff a^k \leq n < (a+1)^k$,
so there are exactly $(a+1)^k - a^k$ different integers such that $\lfloor \sqrt[k]{n} \rfloor = a$

$$\begin{aligned}&= \alpha_0 A \left(\sum_{n=0}^{\infty} \sum_{i=0}^{k-1} \binom{k}{i} n^i A^n \right) \mathbb{1}, \quad \text{by using the binomial theorem} \\ &= \alpha_0 A \left(\sum_{i=0}^{k-1} \binom{k}{i} \sum_{n=0}^{\infty} n^i A^n \right) \mathbb{1}, \quad \text{by changing the order of summation} \\ &= \alpha_0 A \left(\sum_{i=0}^{k-1} \binom{k}{i} (I-A)^{-1} \sum_{j=0}^i j! \left\{ \begin{matrix} i \\ j \end{matrix} \right\} (A(I-A)^{-1})^j \right) \mathbb{1}, \quad \text{by Equation (4.7)} \\ &= \alpha_0 A (I-A)^{-1} \left(\sum_{i=0}^{k-1} \sum_{j=0}^i j! \binom{k}{i} \left\{ \begin{matrix} i \\ j \end{matrix} \right\} (A(I-A)^{-1})^j \right) \mathbb{1} \\ &= \alpha_0 A (I-A)^{-1} \left(\sum_{j=0}^{k-1} \sum_{i=j}^{k-1} j! \binom{k}{i} \left\{ \begin{matrix} i \\ j \end{matrix} \right\} (A(I-A)^{-1})^j \right) \mathbb{1}, \quad \text{by changing the order of summation} \\ &= \alpha_0 A (I-A)^{-1} \left(\sum_{j=0}^{k-1} j! (A(I-A)^{-1})^j \sum_{i=j}^{k-1} \binom{k}{i} \left\{ \begin{matrix} i \\ j \end{matrix} \right\} \right) \mathbb{1} \\ &= \alpha_0 A (I-A)^{-1} \left(\sum_{j=0}^{k-1} j! (A(I-A)^{-1})^j \left(\left\{ \begin{matrix} k+1 \\ j+1 \end{matrix} \right\} - \left\{ \begin{matrix} k \\ j \end{matrix} \right\} \right) \right) \mathbb{1}, \quad \text{by Equation (4.3)} \\ &= \alpha_0 A (I-A)^{-1} \left(\sum_{j=0}^{k-1} j! (A(I-A)^{-1})^j (j+1) \left\{ \begin{matrix} k \\ j+1 \end{matrix} \right\} \right) \mathbb{1}, \quad \text{by Equation (4.5)} \\ &= \alpha_0 A (I-A)^{-1} \left(\sum_{j=0}^{k-1} (j+1)! \left\{ \begin{matrix} k \\ j+1 \end{matrix} \right\} (A(I-A)^{-1})^j \right) \mathbb{1} \\ &= \alpha_0 \left(\sum_{j=1}^k j! \left\{ \begin{matrix} k \\ j \end{matrix} \right\} (A(I-A)^{-1})^j \right) \mathbb{1}.\end{aligned}$$

Lastly, we check that r 's distribution satisfies the moments problem. This is equivalent to checking that the moment generating function of r 's distribution has a non-zero radius around the origin [29]. Indeed, for all $s \in \mathbb{R}$:

$$\begin{aligned}
M_r(s) &= \mathbb{E}(e^{sr}) \\
&= \sum_{n=0}^{\infty} e^{sn} \mathbb{P}(r = n) \\
&= \sum_{n=0}^{\infty} e^{sn} \alpha_0 (P^* I_R)^n P^* I_D \mathbb{1} \\
&= \alpha_0 \left(\sum_{n=0}^{\infty} (e^s P^* I_R)^n \right) P^* I_D \mathbb{1},
\end{aligned}$$

where this last series is geometric; and in fact, within its interval of convergence we have:

$$M_r(s) = \alpha_0 (I - e^s P^* I_R)^{-1} P^* I_D \mathbb{1}.$$

Convergence of this matrix geometric series is given by the values of $s \in \mathbb{R}$ such that

$$\rho(e^s P^* I_R) = e^s \rho(P^* I_R) < 1 \iff s < -\log(\rho(P^* I_R)),$$

and since $\rho(P^* I_R) < 1$, we have $-\log \rho(P^* I_R) > 0$, so the series converges in $(-\infty, -\log \rho(P^* I_R))$, which is an interval around the origin, as we wanted to show. $\square$